PE-EC801B: Fiber Optic Communication
Study Guide and Comprehensive Exam Preparation Notes

Semester VIII B.Tech ECE (MAKAUT)
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Chapter 1

Introduction to Optical Fiber Communication

1.1 Section 1: Basics of Optical Communication [10M] [Priority: High]

1.1.1 1. General vs. Fiber Optic Communication Systems

A communication system transmits information from a source to a destination. In an optical communication system, light is used as the information carrier, and the transmission medium is an optical fiber cable.

A. General Communication System Block Diagram

A standard communication system consists of three main elements: Transmitter, Channel, and Receiver.

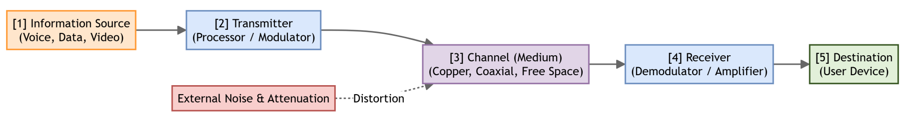

Figure 1.1: General Communication System Block Diagram

1. **Information Source:** Generates the input message signal (analog voice, digital data, video).

2. **Transmitter:** Processes and modulates the source signal onto a suitable carrier wave (RF, microwave) for transmission.

3. **Channel (Medium):** The physical path over which the modulated signal propagates (free space, copper wire, coaxial cable). Noise and attenuation degrade the signal here.

4. **Receiver:** Demodulates and amplifies the received signal, converting it back to the original message format.

5. **Destination:** The final recipient of the transmitted message.

B. Fiber Optic Communication System Block Diagram

An optical fiber communication system adapts the general framework to use light signals and optical fibers.

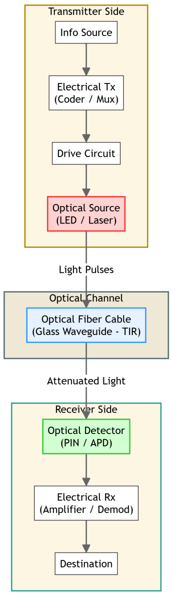

Figure 1.2: Fiber Optic Communication System Block Diagram

1. **Information Source & Electrical Transmitter:** Converts non-electrical messages (like voice) into electrical signals and processes them (coding, multiplexing).
2. **Drive Circuit:** Modulates the electrical signal into a drive current suitable for operating the optical source.
3. **Optical Source:** Converts electrical signal variations into light pulses.

LED (Light Emitting Diode):* Used for low-speed, short-distance transmission. Laser Diode:* Used for high-speed, long-distance transmission.

1. **Optical Fiber Cable (Channel):** A cylindrical glass/plastic waveguide that guides light pulses via **Total Internal Reflection (TIR)**.
2. **Optical Detector (Photodetector):** Converts the received light pulses back into electrical current.

PIN Photodiode:* Simple, low noise, no internal gain. APD (Avalanche Photodiode):* High sensitivity, high response speed, possesses internal multiplication gain.

1. **Electrical Receiver (Amplifier/Demodulator):** Amplifies the weak photodetector current, filters out noise, and reconstructs the original signal.
2. **Destination:** The end-user device or computer system.

1.1.2 2. Advantages & Disadvantages of Optical Fiber Communication

A. Advantages

- **Enormous Bandwidth:** Light frequencies are in the terahertz range (10^{14} Hz), providing a transmission bandwidth-distance product thousands of times greater than copper cables or microwave links.
- **Low Attenuation (Transmission Loss):** Modern silica glass fibers have losses as low as 0.2 dB/km at $\lambda = 1550$ nm, allowing signal transmission over 100 km without repeaters.
- **Immunity to Electromagnetic Interference (EMI):** Being made of dielectric materials (glass/plastic), fibers are completely unaffected by external magnetic fields, lightning strikes, high-voltage lines, or radio frequency signals.
- **Small Size and Lightweight:** An optical fiber has a core/cladding diameter of about 125 μm, making cables much thinner and lighter than copper bundles.
- **High Security & No Crosstalk:** Light is confined entirely within the fiber core. There is no electromagnetic radiation escaping the cable, making unauthorized tapping virtually impossible and eliminating crosstalk between adjacent fibers.
- **Electrical Isolation:** Since fibers are non-conductive, there is no risk of electrical shocks, short circuits, or ground loops, making them safe for hazardous environments (chemical plants, oil refineries).

B. Disadvantages

- **High Initial Installation Cost:** The cost of specialized optical transmitters, receivers, and deployment equipment is higher than copper equivalents.
- **Fragility and Splicing Difficulty:** Glass fibers are fragile and susceptible to damage under bending stress. Splicing (joining) two fibers requires expensive fusion splicing tools and highly skilled technicians.
- **Specialized Components:** Optical connectors, splitters, and amplifiers are complex to design and align.
- **Cannot Carry Electrical Power:** Unlike copper wires, optical fibers cannot transmit electrical power directly to remote equipment (e.g., telephone sets, sensors).

1.2 Section 2: Ray Optics & Light Propagation Principles [15M] [Priority: High]

Light propagation in large-core fibers ($d \gg \lambda$) is modeled using **Geometric (Ray) Optics**, assuming light travels in straight lines (rays) and changes direction at interfaces between different media.

1.2.1 1. Refraction, Snell's Law, Critical Angle, and Total Internal Reflection (TIR)

- **Refraction:** The bending of a light ray as it passes from one transparent medium with refractive index n_1 to another with refractive index n_2, caused by the change in the speed of light.

- **Snell's Law:** Governs refraction at the interface of two media:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

where θ_1 is the angle of incidence and θ_2 is the angle of refraction.

- **Critical Angle (θ_c):** When light travels from a denser medium (n_1) to a rarer medium (n_2, where $n_1 > n_2$), the refracted ray bends away from the normal. As the angle of incidence (θ_1) increases, the angle of refraction (θ_2) reaches 90° (propagating along the interface). The angle of incidence that causes $\theta_2 = 90°$ is the **Critical Angle (θ_c)**:

$$n_1 \sin(\theta_c) = n_2 \sin(90°) \implies \theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

- **Total Internal Reflection (TIR):** The phenomenon where a light ray traveling in a denser medium hits the interface with a rarer medium and is entirely reflected back into the denser medium.

Conditions for TIR:*

1. The light ray must travel from a medium of higher refractive index (n_1) to a medium of lower refractive index (n_2) ($n_1 > n_2$).
2. The angle of incidence at the interface must exceed the critical angle ($\theta_i > \theta_c$).

1.2.2 2. Acceptance Angle & Numerical Aperture (NA) Derivation

The **Acceptance Angle (θ_a)** is the maximum angle of incidence at the air-core interface for which the light ray will be totally internally reflected at the core-cladding interface and thus propagate along the fiber. The **Numerical Aperture (NA)** represents the light-gathering capacity of the fiber.

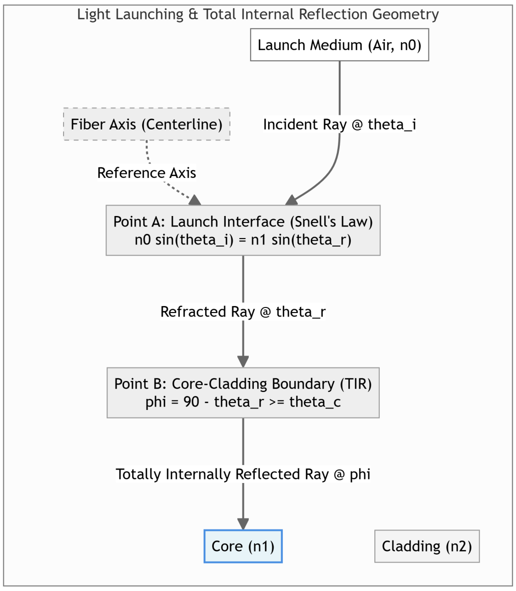

Figure 1.3: Ray Propagation & Acceptance Angle

Step-by-Step Derivation:

1. Let n_0 be the refractive index of the launching medium (usually air, $n_0 = 1$), n_1 be the core refractive index, and n_2 be the cladding refractive index ($n_1 > n_2$).

2. Let a light ray enter the fiber core from the launching medium at an angle of incidence θ_i with the fiber axis. Let θ_r be the angle of refraction inside the core.

3. By **Snell's Law** at the air-core interface (Point A):

$$n_0 \sin(\theta_i) = n_1 \sin(\theta_r) \quad \text{— (Eq 1.1)}$$

1. The refracted ray travels through the core and strikes the core-cladding interface (Point B) at an angle of incidence ϕ. From the geometry of the right-angled triangle ABC:

$$\phi = 90^\circ - \theta_r \quad \text{— (Eq 1.2)}$$

1. For the ray to undergo Total Internal Reflection (TIR) at Point B, we must have:

$$\phi \geq \theta_c \implies \sin(\phi) \geq \sin(\theta_c)$$

1. Since $\sin(\theta_c) = \frac{n_2}{n_1}$, this condition becomes:

$$\sin(\phi) \geq \frac{n_2}{n_1} \quad \text{— (Eq 1.3)}$$

1. Using Eq 1.2, we rewrite $\sin(\phi)$ as:

$$\sin(\phi) = \sin(90^\circ - \theta_r) = \cos(\theta_r)$$

1. Substituting this into Eq 1.3:

$$\cos(\theta_r) \geq \frac{n_2}{n_1}$$

1. Using the trigonometric identity $\sin(\theta_r) = \sqrt{1-\cos^2(\theta_r)}$:

$$\sin(\theta_r) \leq \sqrt{1-\left(\frac{n_2}{n_1}\right)^2} = \sqrt{\frac{n_1^2-n_2^2}{n_1^2}} = \frac{\sqrt{n_1^2-n_2^2}}{n_1}$$

1. Substitute this inequality back into Snell's Law (Eq 1.1):

$$n_0\sin(\theta_i) \leq n_1\left(\frac{\sqrt{n_1^2-n_2^2}}{n_1}\right) = \sqrt{n_1^2-n_2^2}$$

1. The maximum angle of incidence $\theta_i = \theta_a$ (Acceptance Angle) occurs at the limit:

$$\sin(\theta_a) = \frac{\sqrt{n_1^2-n_2^2}}{n_0}$$

1. For air launching ($n_0 = 1$), the **Acceptance Angle (θ_a)** is:

$$\theta_a = \sin^{-1}\left(\sqrt{n_1^2-n_2^2}\right)$$

1. The **Numerical Aperture (NA)** is defined as the sine of the acceptance angle in air:

$$\mathrm{NA} = \sin(\theta_a) = \sqrt{n_1^2-n_2^2}$$

1. Using the **Relative Refractive Index Difference** (Δ):

$$\Delta = \frac{n_1^2-n_2^2}{2n_1^2} \approx \frac{n_1-n_2}{n_1}$$

We can rewrite the expression as:

$$n_1^2-n_2^2 = 2n_1^2\Delta \implies \mathrm{NA} = n_1\sqrt{2\Delta}$$

1.2.3 3. Meridional vs. Skew Rays

Light rays propagating along an optical fiber are classified based on their path and intersection with the fiber's central axis.

Feature / Criteria	Meridional Rays	Skew Rays
Path Definition	Pass through the fiber's longitudinal axis.	Do not pass through the fiber's longitudinal axis.
Propagation Geometry	Confined to a single plane containing the axis (meridional plane).	Follow a helical or spiral path around the axis.
Mathematical Analysis	Simple; modeled using 2D geometry and ray tracing.	Complex; requires 3D cylindrical geometry.
Acceptance Condition	Determined by the standard NA equation: $\sin(\theta_a) = \mathrm{NA}$.	Can be accepted at larger launch angles than meridional rays.
Total Internal Reflection	Occurs at a flat cylindrical surface point.	Occurs along a helical trajectory path at the core boundary.

1.3 Section 3: Electromagnetic Model & Vector Nature of Light [5M] [Priority: Medium]

When the core diameter (d) of the fiber is comparable to the wavelength of the light (λ), geometric optics fails. In this regime, light must be modeled as electromagnetic waves using Maxwell's equations.

1.3.1 1. Ray Model vs. Wave Model

Feature / Criteria	Ray Model (Geometric Optics)	Wave Model (Waveguide Theory)
Light Representation	Discrete light rays traveling in straight lines.	Continuous electromagnetic waves (E & H fields).
Wavelength Assumption	Assumes wavelength is negligible ($\lambda \to 0$).	Accounts for finite wavelength (λ).
Fiber Compatibility	Valid only for multimode fibers ($d \gg \lambda$).	Mandatory for single-mode fibers ($d \approx \lambda$).
Target Phenomena	Describes refraction, reflection, critical angle, and NA.	Explains diffraction, interference, modal cutoff, and dispersion.
Resulting Concept	Light paths (continuous geometric angles).	Discrete electromagnetic modes (guided wave patterns).

1.3.2 2. Vector Nature of Light & cylindrical Dielectric Rod

Light waves consist of oscillating electric ($\vec{E}$) and magnetic ($\vec{H}$) fields, which are vectors perpendicular to each other and to the direction of propagation.

- **Maxwell's Equations:** Light propagation in a cylindrical dielectric rod (representing the fiber core) is modeled by solving the electromagnetic wave equations in cylindrical coordinates (r, ϕ, z):

$$\nabla^2 \vec{E} - n^2 \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

- **Modes of Propagation:** Because of the vector boundary conditions at the cylindrical core-cladding interface, the electromagnetic waves cannot propagate as simple transverse electromagnetic (TEM) waves. Instead, they form discrete modes:

1. **Transverse Electric (TE) Modes:** No electric field component in the direction of propagation ($E_z = 0$).
2. **Transverse Magnetic (TM) Modes:** No magnetic field component in the direction of propagation ($H_z = 0$).
3. **Hybrid Modes (HE and EH):** Both electric and magnetic fields have non-zero components in the direction of propagation ($E_z \neq 0, H_z \neq 0$). These arise because of skew rays and the cylindrical boundary curvature.

1.4 Section 4: Practice Problems [10M] [Priority: High]

1.4.1 Problem 1.1: Critical Angle Calculation

Given:

- Core refractive index (n_1) = **1.82**
- Cladding refractive index (n_2) = **1.73**

Formula:

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

Substitution & Calculation:

$$\theta_c = \sin^{-1}\left(\frac{1.73}{1.82}\right)$$
$$\theta_c = \sin^{-1}(0.95055)$$
$$\theta_c \approx 71.91^\circ$$

Final Answer: The critical angle at the core-cladding interface is 71.91°.

1.4.2 Problem 1.2: Numerical Aperture & Index Tuning

Given:

- Core refractive index (n_1) = **1.48**
- Cladding refractive index (n_2) = **1.46**
- Target NA = **0.23**

Formulas:

1. $\mathrm{NA} = \sqrt{n_1^2 - n_2^2}$
2. $n_2' = \sqrt{n_1^2 - (\mathrm{NA}')^2}$

Step 1: Calculate the initial NA:

$$\mathrm{NA} = \sqrt{(1.48)^2 - (1.46)^2}$$
$$\mathrm{NA} = \sqrt{2.1904 - 2.1316} = \sqrt{0.0588}$$
$$\mathrm{NA} \approx 0.242$$

Step 2: Calculate the new cladding index (n_2') for target NA $= 0.23$:

$$n_2' = \sqrt{(1.48)^2 - (0.23)^2}$$
$$n_2' = \sqrt{2.1904 - 0.0529} = \sqrt{2.1375}$$
$$n_2' \approx 1.462$$

Final Answer:

- The initial Numerical Aperture is **0.242**.
- The new cladding refractive index (n_2') required to change the NA to 0.23 is **1.462**.

1.4.3 Problem 1.3: Acceptance Angle in Air

Given:

- Core refractive index (n_1) = **1.50**
- Cladding refractive index (n_2) = **1.47**
- Launch medium = Air ($n_0 = 1$)

Formulas:

1. Critical Angle: $\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$
2. Numerical Aperture: $\text{NA} = \sqrt{n_1^2 - n_2^2}$
3. Acceptance Angle: $\theta_a = \sin^{-1}(\text{NA})$

Substitution & Calculation:

1. **Critical Angle (θ_c):**

$$\theta_c = \sin^{-1}\left(\frac{1.47}{1.50}\right) = \sin^{-1}(0.98)$$
$$\theta_c \approx 78.52^\circ$$

1. **Numerical Aperture (NA):**

$$\text{NA} = \sqrt{(1.50)^2 - (1.47)^2} = \sqrt{2.25 - 2.1609} = \sqrt{0.0891}$$
$$\text{NA} \approx 0.2985$$

1. **Acceptance Angle (θ_a):**

$$\theta_a = \sin^{-1}(0.2985)$$
$$\theta_a \approx 17.37^\circ$$

Final Answer:

- The critical angle is 78.52°.
- The Numerical Aperture is **0.2985**.
- The acceptance angle in air is 17.37°.

Chapter 2

Optical Fibers & Waveguide Theory

2.1 Section 1: Fiber Classification & Index Profiles [10M] [Priority: High]

Optical fibers are categorized according to their refractive index profiles and the number of electromagnetic modes they propagate.

2.1.1 1. Step-Index (SI) vs. Graded-Index (GI) Fibers

- **Step-Index (SI) Fiber:** The core has a uniform refractive index (n_1) throughout, which drops abruptly (in a "step") to a lower cladding refractive index (n_2) at the core-cladding boundary.

- **Graded-Index (GI) Fiber:** The refractive index of the core decreases continuously in a parabolic profile from the center (n_1) out to the cladding boundary (n_2). The profile is governed by:

$$n(r) = \begin{cases} n_1\sqrt{1 - 2\Delta\left(\frac{r}{a}\right)^{\alpha}} & \text{for } r < a \\ n_2 = n_1\sqrt{1-2\Delta} & \text{for } r \geq a \end{cases}$$

where a is the core radius, α is the profile parameter ($\alpha \approx 2$ for a parabolic profile), and r is the radial distance.

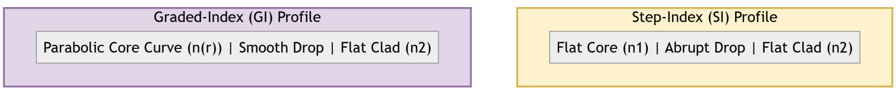

Figure 2.1: Refractive Index Profiles

Feature / Criteria	Step-Index (SI) Fiber	Graded-Index (GI) Fiber
Index Profile	Abrupt step at core-cladding boundary.	Continuous, parabolic decrease from center.
Light Propagation	Zig-zag paths via discrete reflections (TIR).	Continuous sinusoidal, curving refraction paths.
Intermodal Dispersion	High; rays travel different lengths, causing pulse spread.	Low; helical rays travel faster in outer regions, self-focusing.
Bandwidth	Low (typically **20 MHz·km** for multimode).	High (typically **2 GHz·km** for multimode).
Numerical Aperture	Constant across the core: NA $= \sqrt{n_1^2 - n_2^2}$.	Maximum at axis, decreases to zero at cladding boundary.
Source Coupling	Easy; large core and constant NA.	Moderately difficult; requires precise alignment.

2.1.2 2. Single-Mode (SMF) vs. Multimode (MMF) Fibers

- **Single-Mode Fiber (SMF):** Has a small core diameter (8–10 μm) which permits only the fundamental mode (HE_{11}) to propagate at the operating wavelength.
- **Multimode Fiber (MMF):** Has a larger core diameter (50–100 μm) which allows many hundreds of different electromagnetic modes to propagate simultaneously.

Feature / Criteria	Single-Mode Fiber (SMF)	Multimode Fiber (MMF)
Core Diameter	Very small (8–10 μm).	Large (50–100 μm).
Mode Count	Exactly 1 mode (HE_{11}).	Hundreds of modes ($N \gg 100$).
Intermodal Dispersion	Zero (only one mode exists).	High (significant mode delay differences).
Compatible Sources	Laser Diodes (LD) only.	LEDs and Laser Diodes.
Attenuation / Losses	Extremely low (0.2 dB/km at 1550 nm).	Moderately high (1.0–3.0 dB/km).
Applications	Long-haul telecom, undersea cables.	Short-haul LANs, data centers, building backbones.

2.2 Section 2: Key Waveguide Parameters [5M] [Priority: High]

2.2.1 1. Normalized Frequency (V-number)

The **V-number** is a dimensionless parameter that determines the number of modes supported by a step-index fiber and dictates its wave propagation limits.

- **Equation:**

$$V = \frac{2\pi a}{\lambda}\text{NA} = \frac{2\pi a}{\lambda}\sqrt{n_1^2 - n_2^2}$$

where a is the core radius, and λ is the operating wavelength in free space.

- **Single-Mode Propagation Condition:** A step-index fiber propagates only the fundamental HE_{11} mode if the V-number is less than the first zero of the Bessel function J_0:

$$V < 2.405$$

- **Guided Modes Count (N):**
- For Step-Index Multimode Fibers ($V \gg 2.405$):

$$N_{SI} \approx \frac{V^2}{2}$$

- For Graded-Index Parabolic Fibers ($\alpha = 2$):

$$N_{GI} \approx \frac{V^2}{4}$$

(GI fibers support half the number of modes as SI fibers for the same V-number).

2.2.2 2. Cutoff Wavelength (λ_c)

The **Cutoff Wavelength** is the transition wavelength above which the fiber supports only a single mode (HE_{11}). Below λ_c, higher-order modes begin to propagate, and the fiber becomes multimode.

- **Equation:**

$$\lambda_c = \frac{2\pi a\text{NA}}{V_c} = \frac{2\pi a\sqrt{n_1^2 - n_2^2}}{2.405}$$

where $V_c = 2.405$ is the single-mode cutoff limit.

2.2.3 3. Mode-Field Diameter (MFD)

In single-mode fibers, the light is not entirely confined within the core; a fraction of the optical power propagates as an evanescent wave inside the cladding. The **Mode-Field Diameter (MFD)** is a measure of the radial distribution of the optical power of the fundamental mode.

- **Definition:** The radial distance between the points where the optical power density drops to $1/e^2$ (or electric field strength drops to $1/e$) of its maximum value at the fiber axis.
- **Marcuse Formula (for Step-Index Fibers):**

$$\frac{W}{a} \approx 0.65 + 1.619V^{-1.5} + 2.879V^{-6}$$

where W is the mode-field radius ($MFD = 2W$), and a is the core radius.

- **Significance:** A smaller MFD increases light confinement but makes splice alignment more sensitive and increases susceptibility to nonlinear effects.

2.3 Section 3: Waveguide Mode Theory & Modal Analysis [15M] [Priority: High]

To describe light propagation in fibers of sub-wavelength dimensions, we solve the vector wave equation derived from Maxwell's equations under cylindrical boundary conditions.

2.3.1 1. Maxwell's Equations in Cylindrical Coordinates

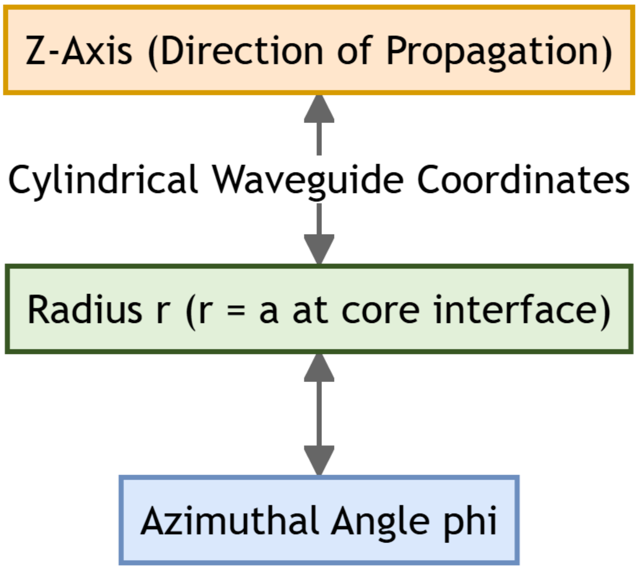

Figure 2.2: Cylindrical Coordinate Boundaries

In cylindrical coordinates (r, ϕ, z), assuming waves propagate in the z-direction with propagation constant β and time-harmonic dependence $e^{j(\omega t - \beta z)}$, the electric $(\vec{E})$ and magnetic $(\vec{H})$ field vectors are:

$$\vec{E} = \vec{E}_0(r, \phi) e^{j(\omega t - \beta z)}, \quad \vec{H} = \vec{H}_0(r, \phi) e^{j(\omega t - \beta z)}$$

Maxwell's curl equations ($\nabla \times \vec{E} = -j\omega\mu\vec{H}$ and $\nabla \times \vec{H} = j\omega\varepsilon\vec{E}$) yield the following component equations:

1. $\frac{1}{r}\frac{\partial E_z}{\partial \phi} + j\beta E_\phi = -j\omega\mu H_r$
2. $-j\beta E_r - \frac{\partial E_z}{\partial r} = -j\omega\mu H_\phi$
3. $\frac{1}{r}\left[\frac{\partial}{\partial r}(rE_\phi) - \frac{\partial E_r}{\partial \phi}\right] = -j\omega\mu H_z$
4. $\frac{1}{r}\frac{\partial H_z}{\partial \phi} + j\beta H_\phi = j\omega\varepsilon E_r$
5. $-j\beta H_r - \frac{\partial H_z}{\partial r} = j\omega\varepsilon E_\phi$
6. $\frac{1}{r}\left[\frac{\partial}{\partial r}(rH_\phi) - \frac{\partial H_r}{\partial \phi}\right] = j\omega\varepsilon E_z$

2.3.2 2. Transverse Fields in Terms of Longitudinal Fields

By algebraic manipulation of components 1, 2, 4, and 5, the transverse field components $(E_r, E_\phi, H_r, H_\phi)$ can be expressed entirely in terms of the longitudinal field components (E_z, H_z):

$$E_r = \frac{-j}{q^2}\left[\beta\frac{\partial E_z}{\partial r} + \frac{\omega\mu}{r}\frac{\partial H_z}{\partial \phi}\right] \quad \text{— (Eq 2.1)}$$

$$E_\phi = \frac{-j}{q^2}\left[\frac{\beta}{r}\frac{\partial E_z}{\partial \phi} - \omega\mu\frac{\partial H_z}{\partial r}\right] \quad \text{— (Eq 2.2)}$$

$$H_r = \frac{-j}{q^2}\left[\beta\frac{\partial H_z}{\partial r} - \frac{\omega\varepsilon}{r}\frac{\partial E_z}{\partial \phi}\right] \quad \text{— (Eq 2.3)}$$

$$H_\phi = \frac{-j}{q^2}\left[\frac{\beta}{r}\frac{\partial H_z}{\partial \phi} + \omega\varepsilon\frac{\partial E_z}{\partial r}\right] \quad \text{— (Eq 2.4)}$$

where $q^2 = k^2n^2 - \beta^2 = \omega^2\mu\varepsilon - \beta^2$.

This reduction simplifies the waveguide analysis to solving a scalar wave equation for E_z and H_z and applying boundary conditions.

2.3.3 3. Helmholtz Wave Equation & Bessel Solutions

The wave equation for the longitudinal components E_z (and similarly for H_z) is:

$$\frac{\partial^2 E_z}{\partial r^2} + \frac{1}{r}\frac{\partial E_z}{\partial r} + \frac{1}{r^2}\frac{\partial^2 E_z}{\partial \phi^2} + q^2 E_z = 0$$

Using separation of variables, $E_z(r,\phi) = F(r)e^{j\nu\phi}$ (where ν is an integer representing azimuthal variation), we obtain the **Bessel Differential Equation** for the radial function $F(r)$:

$$\frac{d^2F}{dr^2} + \frac{1}{r}\frac{dF}{dr} + \left(q^2 - \frac{\nu^2}{r^2}\right)F = 0$$

Solutions:

1. **Core Region ($r < a$, index n_1):**

The transverse parameter is $u^2 = a^2(k_0^2n_1^2 - \beta^2) > 0$. The solution must remain finite at $r = 0$, which requires the **Bessel function of the first kind (J_ν)**:

$$E_z(r,\phi) = AJ_\nu\left(\frac{ur}{a}\right)e^{j\nu\phi}, \quad H_z(r,\phi) = BJ_\nu\left(\frac{ur}{a}\right)e^{j\nu\phi}$$

1. **Cladding Region ($r > a$, index n_2):**

For guided waves, the fields must decay to zero as $r \to \infty$. The transverse parameter is $w^2 = a^2(\beta^2 - k_0^2n_2^2) > 0$. The solution requires the **Modified Bessel function of the second kind (K_ν)**:

$$E_z(r,\phi) = CK_\nu\left(\frac{wr}{a}\right)e^{j\nu\phi}, \quad H_z(r,\phi) = DK_\nu\left(\frac{wr}{a}\right)e^{j\nu\phi}$$

2.3.4 4. Boundary Conditions & Eigenvalue Equation

At the core-cladding interface ($r = a$), the tangential components of the electric and magnetic fields must be continuous:

$$E_z(a^-) = E_z(a^+), \quad H_z(a^-) = H_z(a^+)$$

$$E_\phi(a^-) = E_\phi(a^+), \quad H_\phi(a^-) = H_\phi(a^+)$$

Substituting the Bessel solutions for E_z and H_z into Equations 2.2 and 2.4 to find E_ϕ and H_ϕ, and matching them at $r = a$ yields a system of four linear equations for constants A, B, C, and D. Setting the determinant of this system to zero gives the transcendental **Characteristic Equation**:

$$\left[\frac{J'_\nu(u)}{uJ_\nu(u)} + \frac{K'_\nu(w)}{wK_\nu(w)}\right]\left[\frac{J'_\nu(u)}{uJ_\nu(u)} + \frac{n_2^2}{n_1^2}\frac{K'_\nu(w)}{wK_\nu(w)}\right] = \nu^2\left(\frac{1}{u^2} + \frac{1}{w^2}\right)\left(\frac{1}{u^2} + \frac{n_2^2}{n_1^2}\frac{1}{w^2}\right)$$

- For $\nu = 0$, the right-hand side is zero. The equations decouple, yielding:
- **TE Modes ($\nu = 0, E_z = 0$):** $\frac{J'_0(u)}{uJ_0(u)} + \frac{K'_0(w)}{wK_0(w)} = 0$
- **TM Modes ($\nu = 0, H_z = 0$):** $\frac{J'_0(u)}{uJ_0(u)} + \frac{n_2^2}{n_1^2}\frac{K'_0(w)}{wK_0(w)} = 0$
- For $\nu \neq 0$, the fields are coupled, producing hybrid **HE** and **EH** modes.

2.4 Section 4: Practice Problems [10M] [Priority: High]

2.4.1 Problem 2.1: Multimode Step-Index Fiber Characterization

Given:

- Core diameter ($2a$) = 80 μm $\implies$ core radius (a) = 40 $\mu\text{m} = 40 \times 10^{-6}$ m
- Core refractive index (n_1) = **1.48**
- Relative refractive index difference (Δ) = 1.5% $= 0.015$
- Operating wavelength (λ) = 0.85 $\mu\text{m} = 0.85 \times 10^{-6}$ m

Formulas:

1. Numerical Aperture: $\text{NA} = n_1\sqrt{2\Delta}$
2. Normalized Frequency: $V = \frac{2\pi a}{\lambda}\text{NA}$
3. Total Guided Modes: $N \approx \frac{V^2}{2}$

Step 1: Calculate the Numerical Aperture (NA):

$$\text{NA} = 1.48 \times \sqrt{2 \times 0.015} = 1.48 \times \sqrt{0.03}$$

$$\text{NA} = 1.48 \times 0.1732 = 0.2563$$

Step 2: Calculate the V-number:

$$V = \frac{2\pi \times 40 \times 10^{-6}}{0.85 \times 10^{-6}} \times 0.2563$$

$$V = \frac{251.327}{0.85} \times 0.2563 \approx 295.68 \times 0.2563$$

$$V \approx 75.78$$

Step 3: Calculate the number of guided modes (N):

$$N = \frac{(75.78)^2}{2} = \frac{5742.6}{2} \approx 2871$$

Final Answer:

- The Normalized Frequency (V-number) is **75.78**.
- The total number of guided modes is **2871**.

2.4.2 Problem 2.2: Cutoff Wavelength of Single-Mode Fiber

Given:

- Core radius (a) = 4 μm $= 4 \times 10^{-6}$ m
- Core refractive index (n_1) = **1.48**
- Relative refractive index difference (Δ) = 0.2% $= 0.002$
- Cutoff V-number (V_c) = **2.405**

Formulas:

1. Numerical Aperture: $\text{NA} = n_1\sqrt{2\Delta}$
2. Cutoff Wavelength: $\lambda_c = \frac{2\pi a \text{NA}}{V_c}$

Step 1: Calculate the Numerical Aperture (NA):

$$\text{NA} = 1.48 \times \sqrt{2 \times 0.002} = 1.48 \times \sqrt{0.004}$$

$$\text{NA} = 1.48 \times 0.06325 \approx 0.0936$$

Step 2: Calculate the Cutoff Wavelength (λ_c):

$$\lambda_c = \frac{2\pi \times (4 \times 10^{-6}\ \text{m}) \times 0.0936}{2.405}$$

$$\lambda_c = \frac{2.352 \times 10^{-6}}{2.405}\ \text{m}$$

$$\lambda_c \approx 0.978 \times 10^{-6}\ \text{m} = 0.978\ \mu\text{m} = 978\ \text{nm}$$

Final Answer: The cutoff wavelength of the single-mode fiber is 0.978 μ**m** (or 978 **nm**).

Chapter 3

Signal Degradation in Optical Fibers

This chapter details the physical mechanisms responsible for signal degradation in optical fibers: **Attenuation** (power loss) and **Dispersion** (pulse broadening). It also covers fiber fabrication techniques, cabling methods, and optical measurement techniques, including the Optical Time Domain Reflectometer (OTDR).

3.1 1. Attenuation in Optical Fibers

3.1.1 1.1 Definition and Formula

Attenuation is the reduction in optical power as the light signal propagates through the optical fiber. It limits the maximum transmission distance (link reach) between the transmitter and receiver. The attenuation coefficient α (expressed in dB/km) is defined as:

$$\alpha = \frac{10}{L}\log_{10}\left(\frac{P_{in}}{P_{out}}\right)$$

Where:

- P_{in} is the optical power launched at the input of the fiber (W or mW).
- P_{out} is the optical power exiting the fiber at distance L (W or mW).
- L is the length of the optical fiber (km).

Alternatively, the output power can be expressed as:

$$P_{out} = P_{in}10^{-\alpha L/10}$$

3.1.2 1.2 Absorption Losses

Absorption is a material property where optical energy is converted into heat due to molecular resonances and impurities in the glass. It is categorized into two types:

A. Intrinsic Absorption

Intrinsic absorption is caused by the basic constituent materials of the glass (pure silica) and represents the absolute physical limit of transparency.

- **Ultraviolet Absorption Edge:** Caused by electronic transitions of silica molecules from the valence band to the conduction band. The absorption decays exponentially with increasing wavelength and is negligible in the near-infrared region ($\lambda > 0.8\ \mu$m).
- **Infrared Absorption Tail:** Caused by the vibrational resonances of the silicon-oxygen (Si-O) bonds. This absorption peak occurs around 9.0 μm, and its tail extends down to $\approx 1.6\ \mu$m, setting the upper limit of the usable infrared window.

B. Extrinsic Absorption

Extrinsic absorption is caused by impurities trapped in the glass during manufacturing.

- **Transition Metal Ions:** Impurities such as Iron (Fe^{2+}), Copper (Cu^{2+}), Chromium (Cr^{3+}), and Nickel (Ni^{2+}) absorb light. To maintain losses below 1 dB/km, impurity concentrations must be kept under 1 part per billion (ppb).
- **Hydroxyl (OH^-) Ions:** Caused by water vapor trapped in the silica glass. The OH^- bond has a fundamental vibrational absorption at 2.73 μm, with overtones and combinational vibrations appearing as prominent "water peaks" at 950 nm, 1240 nm, and 1380 nm.

3.1.3 1.3 Scattering Losses (Rayleigh Scattering)

Rayleigh scattering is the dominant loss mechanism in the low-loss window (0.8 μm to 1.6 μm).

- **Physical Cause:** Caused by microscopic, sub-wavelength localized density and composition fluctuations frozen into the silica glass lattice when it cools from a molten state. These fluctuations create localized variations in the refractive index that scatter light in all directions.
- **Wavelength Dependence:** The scattering loss coefficient α_{sc} varies inversely with the fourth power of the wavelength:

$$\alpha_{sc} \propto \frac{1}{\lambda^4}$$

Consequence: Doubling the operating wavelength reduces Rayleigh scattering loss by a factor of 16 (2^4).

3.1.4 1.4 Bending Losses

Bending losses occur when the optical fiber is bent, allowing guided modes to radiate power into the cladding.

A. Macrobending Losses

- **Mechanism:** Occurs when the fiber has a visible curvature (bend radius of a few centimeters). Along a curved path, the phase velocity of the cladding evanescent field must exceed the speed of light in the cladding to keep up with the core field. Since this is physically impossible, the portion of the wave exceeding this limit breaks away and radiates into the cladding.
- **Ray-Optics view:** At the outer curve of the bend, the angle of incidence of the propagating ray falls below the critical angle ($\theta_i < \theta_c$), causing the ray to refract out of the core rather than undergo total internal reflection.

B. Microbending Losses

- **Mechanism:** Caused by microscopic, high-frequency axial deviations (roughness) of the fiber axis, typically having amplitudes of a few microns and periods of a few millimeters.
- **Cause:** Caused by lateral mechanical pressures or stresses applied to the fiber during cabling or winding on a spool.
- **Consequence:** These small bends couple power from lower-order guided modes into higher-order modes and radiation modes, which escape into the cladding.

3.1.5 1.5 Cabling of Optical Fibers

Before deployment in the field, fragile glass fibers must be housed within cables. The primary reasons for cabling are:

- **Mechanical Protection:** Glass has high tensile strength but is highly susceptible to micro-cracks and shear fractures. Cables use strength members (e.g., Kevlar, steel wire) to bear tensile loads during pulling.
- **Environmental Shielding:** Cables protect the fiber from moisture, hydrogen gas penetration (which causes absorption), temperature variations, and chemical attack.
- **Microbending Mitigation:** Buffering tubes (loose or tight buffer designs) isolate the fiber from external lateral pressures, maintaining low attenuation.

3.2 2. Optical Transmission Windows

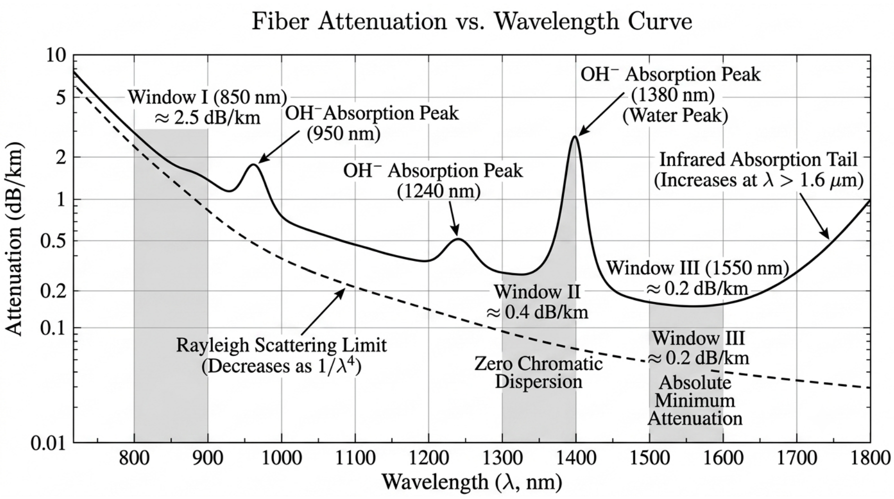

Figure 3.1: Fiber Attenuation vs. Wavelength Curve

Optical fiber communication operates in specific near-infrared spectral regions where silica glass exhibits low loss.

Transmission Window	Wavelength Range	Typical Attenuation	Dispersion Characteristic	Typical Application
First Window	800 to 900 nm	$\approx$ 2.0 to 3.0 dB/km	High chromatic dispersion	Short-reach LANs, GaAs LEDs/VCSELs
Second Window (O-band)	1260 to 1360 nm	$\approx$ 0.4 dB/km	Zero chromatic dispersion	Medium-haul networks, standard SMF
Third Window (C-band)	1530 to 1565 nm	$\approx$ 0.2 dB/km	Minimum loss window, high dispersion	Long-haul, DWDM systems with EDFAs

3.3 3. Dispersion and Pulse Broadening

Dispersion is the temporal spreading of light pulses as they travel along the fiber. If pulses spread too much, they overlap with adjacent pulses, causing **Inter-Symbol Interference (ISI)**. This increases the Bit Error Rate (BER) and limits the bandwidth-distance product of the link.

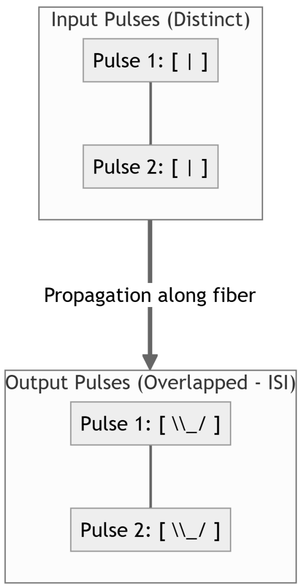

Figure 3.2: Pulse Broadening and Dispersion

3.3.1 3.1 Intermodal Dispersion

Intermodal dispersion occurs only in multimode fibers and is caused by different guided modes traveling at different group velocities.

A. Derivation of Delay Difference (Δt_{modal}) in a Step-Index Multimode Fiber

Consider a step-index fiber of length L with core index n_1 and cladding index n_2.

1. **Fastest Ray:** Travels straight along the fiber axis (z-axis). The time taken t_{min} is:

$$t_{min} = \frac{L}{v_1} = \frac{n_1 L}{c}$$

(where $v_1 = c/n_1$ is the velocity of light in the core).

1. **Slowest Ray:** Propagates at the steepest angle relative to the axis, which is the critical angle of total internal reflection (θ_c). The ray travels a total path length of $L/\sin\theta_c$. The time taken t_{max} is:

$$t_{max} = \frac{L/\sin\theta_c}{v_1} = \frac{n_1 L}{c \sin\theta_c}$$

1. Substitute the critical angle relationship $\sin\theta_c = n_2/n_1$:

$$t_{max} = \frac{n_1^2 L}{n_2 c}$$

1. The total delay difference Δt_{modal} between the slowest and fastest modes is:

$$\Delta t_{modal} = t_{max} - t_{min} = \frac{n_1^2 L}{n_2 c} - \frac{n_1 L}{c} = \frac{n_1 L}{c}\left(\frac{n_1 - n_2}{n_2}\right)$$

1. Using the relative index difference $\Delta \approx \frac{n_1 - n_2}{n_1} \approx \frac{n_1 - n_2}{n_2}$:

$$\Delta t_{modal} \approx \frac{n_1 L}{c}\Delta$$

1. Expressing this in terms of the Numerical Aperture ($\mathrm{NA} \approx n_1\sqrt{2\Delta} \implies \Delta \approx \mathrm{NA}^2/2n_1^2$):

$$\Delta t_{modal} \approx \frac{L(\mathrm{NA})^2}{2 n_1 c}$$

B. RMS Pulse Broadening (σ_{modal})

If we assume the optical power is distributed uniformly across all modes, the root-mean-square (RMS) pulse broadening due to intermodal dispersion is:

$$\sigma_{modal} = \frac{\Delta t_{modal}}{2\sqrt{3}} \approx \frac{n_1 L \Delta}{2\sqrt{3}c}$$

3.3.2 3.2 Intramodal (Chromatic) Dispersion

Intramodal dispersion occurs in all fibers (including single-mode fibers) and is caused by the spectral components of a single mode propagating at different group velocities. The total chromatic dispersion is the sum of material and waveguide dispersion.

$$\tau = D \cdot L \cdot \sigma_\lambda$$

Where:

- τ is the pulse broadening (ps).
- D is the chromatic dispersion coefficient (ps/(nm $\cdot$ km)).
- σ_λ is the spectral width of the light source (nm).

A. Material Dispersion (D_M)

Material dispersion is caused by the wavelength-dependence of the refractive index of the silica glass ($n_1(\lambda)$). Different spectral components of the source see different refractive indices and travel at different velocities.

$$D_M = -\frac{\lambda}{c}\left(\frac{d^2 n_1}{d\lambda^2}\right)$$

B. Waveguide Dispersion (D_W)

Waveguide dispersion is a geometric effect. It arises because the distribution of optical power in a mode between the core and the cladding varies with wavelength.

$$D_W = -\frac{n_2 \Delta}{c\lambda} V \frac{d^2(Vb)}{dV^2}$$

(where b is the normalized propagation constant and V is the normalized frequency).

3.3.3 3.3 Dispersion-Engineered Single-Mode Fibers

By modifying the refractive index profile of the fiber core, the waveguide dispersion D_W (which is typically negative) can be tailored to cancel or shift the material dispersion D_M.

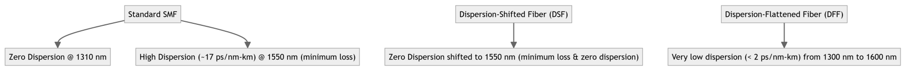

Figure 3.3: Dispersion Curves

- **Dispersion-Shifted Fiber (DSF):** The zero-dispersion wavelength is shifted from 1310 nm to 1550 nm to match the minimum attenuation window. This is achieved by using a triangular index profile to increase the negative waveguide dispersion.
- **Dispersion-Flattened Fiber (DFF):** Uses multiple cladding layers (e.g., quadruple-clad coaxial profile) to keep the total chromatic dispersion extremely low (< 2 ps/(nm $\cdot$ km)) across the entire range from 1300 nm to 1600 nm.

3.4 4. Fiber Fabrication Methods

Fabricating low-loss glass optical fibers requires a two-step process: producing a high-purity glass rod called a **preform**, and then drawing the fiber from this preform in a drawing tower.

3.4.1 4.1 Vapor-Phase Oxidation Techniques

A. Modified Chemical Vapor Deposition (MCVD)

- **Process:** High-purity reactant gases ($SiCl_4, GeCl_4, O_2$) are passed inside a rotating hollow silica tube.
- **Reaction:** An oxygen-hydrogen burner moves back and forth along the tube, heating it externally to $\approx 1600°C$. This causes gas-phase reactions that deposit soot particles on the inner wall of the tube.

- **Sintering:** The burner heats each deposited layer, sintering the soot into a solid, transparent glass layer.
- **Collapse:** Once all core and cladding layers are deposited, the temperature is increased to $\approx 2000°C$, collapsing the hollow tube into a solid glass preform rod.

B. Outside Vapor Deposition (OVD)

- **Process:** Soot particles are deposited on the outside of a rotating mandrel (target rod) using a flame hydrolysis burner.
- **Sintering & Mandrel Removal:** The mandrel is removed after deposition, leaving a porous, hollow soot preform. The preform is then placed in a consolidation furnace, dried with chlorine gas to remove water (OH^- ions), and sintered into a solid glass preform.

C. Vapor Axial Deposition (VAD)

- **Process:** Soot deposition occurs axially on the end of a seed rod rather than radially.
- **Advantage:** Both core and cladding soot are deposited simultaneously using separate burners. The preform is pulled upward and sintered continuously, allowing for the fabrication of very long preforms.

3.4.2 4.2 Double-Crucible Method

- **Process:** A direct melting process that bypasses the preform stage, used primarily for low-melting-point multicomponent glasses.
- **Mechanism:** Core glass is placed in an inner crucible, and cladding glass is placed in a concentric outer crucible. The crucibles are heated, and the molten glasses flow out of concentric orifices at the bottom, forming a cladding-coated glass fiber that is drawn directly onto a spool.

3.5 5. Optical Measurements and OTDR

3.5.1 5.1 Optical Time Domain Reflectometer (OTDR)

An OTDR is an optoelectronic instrument used to characterize an optical fiber. it detects faults, splices, bends, and measures attenuation along the fiber from a single end.

A. Working Principle

The OTDR injects a high-power optical pulse into the fiber and measures the returning optical power as a function of time. The return signal is composed of:

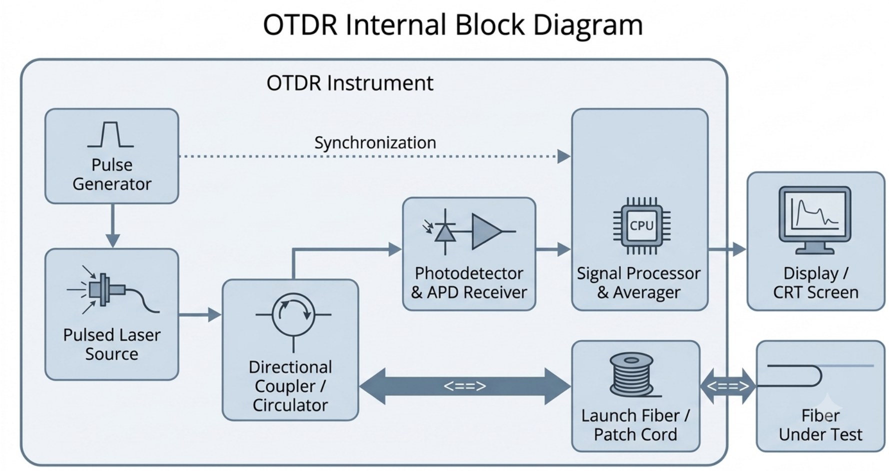

Figure 3.4: OTDR Block Diagram

1. **Rayleigh Backscattering:** Continuous, low-level backscattered light caused by microscopic density fluctuations.
2. **Fresnel Reflections:** Sharp, high-intensity reflections that occur when light encounters abrupt changes in the refractive index (e.g., at connectors, mechanical splices, or cleaved fiber ends). The reflection coefficient is:

$$R = \left(\frac{n_1 - n_{ext}}{n_1 + n_{ext}} \right)^2$$

B. Trace Waveform and Analysis

An OTDR trace plots the backscattered power (in dB) against the distance along the fiber.

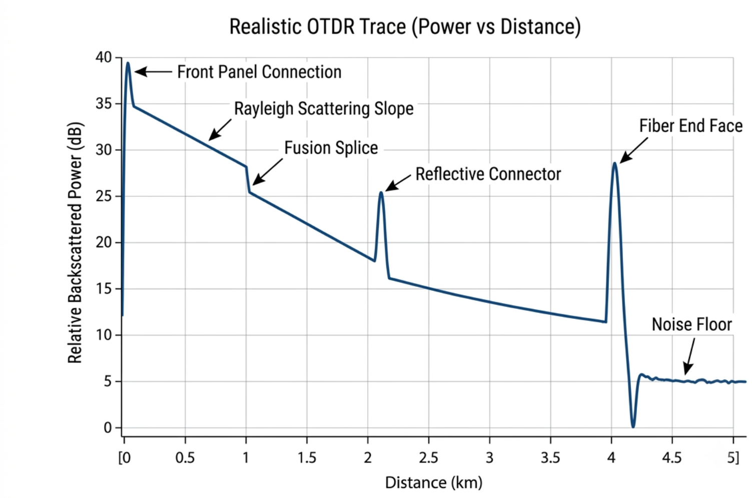

Figure 3.5: OTDR Trace Waveform

- **Rayleigh Slope:** The attenuation coefficient of the fiber is determined from the slope of the linear backscatter regions: Slope $= \Delta P / \Delta z$ dB/km.

- **Non-Reflective Steps:** Indicate localized losses from fusion splices or macrobends.
- **Reflective Peaks:** Indicate connectors, mechanical splices, or physical cracks.
- **End of Fiber:** A final Fresnel peak followed by a drop in power to the receiver noise floor.

3.5.2 5.2 Attenuation Measurement: Cut-Back Method

The Cut-Back method is the standard technique for measuring the attenuation of an optical fiber.

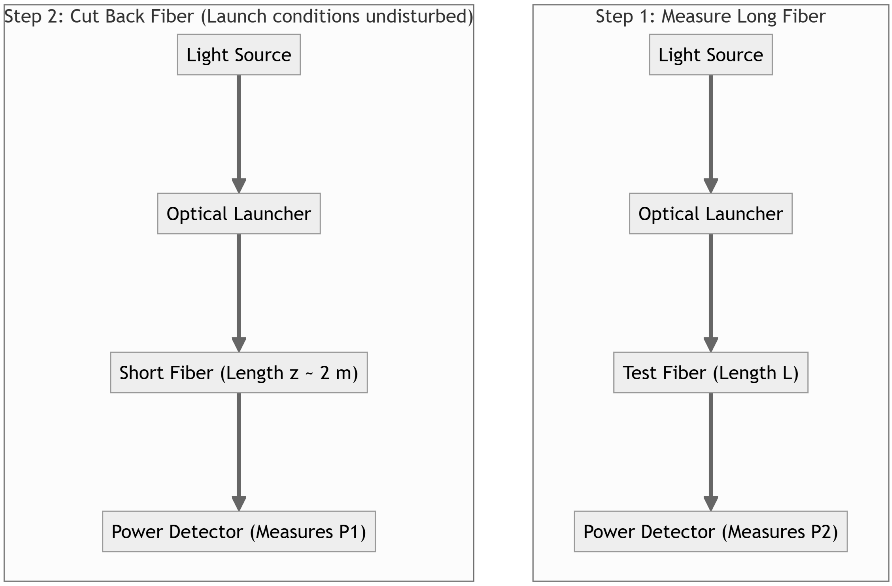

Figure 3.6: Cut-Back Method Setup

1. **Long Fiber Measurement:** Measure the output power $P_2(L)$ at the far end of the test fiber of length L.
2. **Cut-Back:** Without altering the launch conditions at the input end, cut the fiber a few meters from the launcher.
3. **Short Fiber Measurement:** Measure the output power $P_1(z)$ of this short segment (length $z \approx 2$ m).
4. **Calculation:** The attenuation coefficient α is:

$$\alpha = \frac{10}{L-z} \log_{10}\left(\frac{P_1(z)}{P_2(L)}\right) \text{ dB/km}$$

3.5.3 5.3 Numerical Aperture (NA) Far-Field Measurement

The Numerical Aperture is measured by analyzing the far-field radiation pattern exiting the fiber.

1. Light is launched into the fiber, and the exiting beam is projected onto a screen in the far field.

2. A photodetector scans across the radiation pattern at a fixed distance d from the fiber end, recording the optical intensity as a function of the angle θ.

3. The NA is calculated from the half-angle $\theta_{5\%}$ where the output intensity drops to 5% of its peak on-axis value:

$$\mathrm{NA} = \sin(\theta_{5\%})$$

3.6 6. Worked Practice Problems

3.6.1 Problem 6.1: Intermodal Dispersion

An optical link of length $L = 6$ km consists of a multimode step-index fiber. The core refractive index is $n_1 = 1.50$ and the relative index difference is $\Delta = 1.0\%$. Estimate:

1. The intermodal delay difference (Δt_{modal}) between the slowest and fastest modes.

2. The RMS pulse broadening (σ_{modal}) due to intermodal dispersion.

Solution: 1. Given Data:

- $L = 6 \text{ km} = 6 \times 10^3 \text{ m}$
- $n_1 = 1.50$
- $\Delta = 1.0\% = 0.01$
- $c \approx 3 \times 10^8 \text{ m/s}$

2. Formula for Intermodal Delay Difference:

$$\Delta t_{modal} = \frac{n_1 L}{c}\Delta$$

3. Substitution:

$$\Delta t_{modal} = \frac{1.50 \times (6 \times 10^3 \text{ m})}{3 \times 10^8 \text{ m/s}} \times 0.01$$

$$\Delta t_{modal} = \frac{9000}{3 \times 10^8} \times 0.01 = 3 \times 10^{-5} \times 0.01 = 3 \times 10^{-7} \text{ s} = 300 \text{ ns}$$

4. Formula for RMS Pulse Broadening:

$$\sigma_{modal} = \frac{\Delta t_{modal}}{2\sqrt{3}}$$

5. Calculation:

$$\sigma_{modal} = \frac{300 \text{ ns}}{2\sqrt{3}} = \frac{300}{3.464} \approx 86.6 \text{ ns}$$

Final Answer:

1. The intermodal delay difference is 300 **ns**.

2. The RMS pulse broadening is 86.6 **ns**.

3.6.2 Problem 6.2: Attenuation Coefficient

A 2 km length of optical fiber has an input power of $P_{in} = 500\ \mu$W and an output power of $P_{out} = 250\ \mu$W. Calculate the attenuation coefficient of the fiber.

Solution: 1. Given Data:

- $L = 2$ km
- $P_{in} = 500\ \mu$W
- $P_{out} = 250\ \mu$W

2. Formula:

$$\alpha = \frac{10}{L}\log_{10}\left(\frac{P_{in}}{P_{out}}\right)$$

3. Substitution:

$$\alpha = \frac{10}{2}\log_{10}\left(\frac{500}{250}\right)$$

$$\alpha = 5\log_{10}(2)$$

$$\alpha = 5 \times 0.301 = 1.505 \text{ dB/km}$$

Final Answer: The attenuation coefficient of the fiber is 1.51 **dB/km**.

3.7 7. Fiber Connectors, Joints, and GRIN-rod Lenses (Q3.8) [5M][]

Efficiently linking fibers requires specialized mechanical and optical components to minimize coupling losses and maintain alignment.

3.7.1 7.1 GRIN-rod Lenses (Graded-Index Rod Lenses)

A **GRIN-rod lens** is a cylindrical glass rod with a refractive index profile that decreases continuously and parabolically from the central axis outward.

- **Working Principle:** Unlike conventional lenses which use curved glass-air interfaces to refract light, a GRIN-rod lens uses internal index grading to bend light rays continuously along sinusoidal paths.
- **Dimensions:** The majority of GRIN-rod lenses are very compact, with typical diameters in the range of **0.5 to 2 mm**.
- **Alignment Position:** The output face of the transmitting fiber is positioned exactly at the **focal length** (on or near the opposite lens face) of the GRIN-rod lens. This alignment collimates the light, producing a parallel beam with a low divergent angle of between **1 and 5 degrees**, which minimizes longitudinal misalignment effects.

3.7.2 7.2 Lens-Coupled Expanded-Beam Connectors

Expanded-beam connectors use a pair of lenses (often GRIN-rod lenses) to collimate the light exiting the transmitting fiber, letting it cross a small air gap as a wide parallel beam before a second lens focuses it back into the receiving fiber core.

- **Insertion Loss:** Lens-coupled expanded-beam connectors exhibit average losses of **0.7 dB** for both single-mode and graded-index fibers.
- **Divergence Factors:** Several factors cause the collimated beam exiting a GRIN-rod lens to diverge:

1. *Refractive Index Profile:* Non-ideal parabolic index distributions.
2. *Chromatic Aberration:* Different wavelengths focus at slightly different spots.
3. *Size of Fiber Core:* A larger core deviates from a perfect point source at the focal point.

Note on Lens Cut Length:* The **lens cut length** determines the collimation state (e.g. quarter-pitch length for collimation) but is a fixed design property; it is not a dynamic factor causing random beam divergence in an aligned lens.

Chapter 4

Optical Sources & Detectors

This chapter compiles study materials for **Unit IV: Optical Sources & Detectors**. It covers the working principles of optical sources (LEDs and Lasers) and photodetectors (PIN and APD), receiver noise mechanisms, and the design equations for optical links (link power budget, rise-time budget, and the quantum limit).

4.1 1. Optical Sources (LEDs vs. Laser Diodes)

Optical transmitters convert electrical signals into optical signals. The two primary semiconductor light sources used in optical fiber communication are Light Emitting Diodes (LEDs) and Laser Diodes (LDs).

4.1.1 1.1 Comparison Table

Feature / Criteria	Light Emitting Diode (LED)	Laser Diode (LD)
Emission Process	Spontaneous emission (recombination of carriers without external trigger).	Stimulated emission (excited carriers triggered by incident photons).
Light Output	Incoherent light.	Highly coherent light (in-phase wavefronts).
Spectral Width ($\Delta\lambda$)	Broad (30 to 50 nm).	Narrow ($<$ 1 to 3 nm or single-frequency).
Output Power	Low (microwatts to a few milliwatts).	High (tens of milliwatts to watts).
Coupling Efficiency	Low (isotropic, Lambertian emission pattern).	High (directional, well-defined, narrow beam).
Modulation Bandwidth	Lower (up to a few hundred MHz).	Very high (up to tens of GHz).
Temperature Sensitivity	Low.	High (requires thermoelectric cooling).
Lifespan & Reliability	Extremely high, long lifetime.	High, but shorter than LEDs due to high current density.
Typical Applications	Short-reach links, local area networks (LANs).	Long-haul telecom, high-speed DWDM, CATV.

4.1.2 1.2 Population Inversion

In a semiconductor at thermal equilibrium, most electrons reside in the lower energy state (valence band), and the higher energy state (conduction band) is mostly empty.

- **Definition: Population Inversion** is a non-equilibrium state where the population of electrons in the conduction band (higher energy level) exceeds the population of electrons in the valence band (lower energy level).
- **Mechanism:** Achieved by heavily doping the p-type and n-type regions to form a p-n junction, and applying a high forward bias (pumping) that injects a high density of electrons and holes into the thin active region.
- **Significance:** It is the fundamental prerequisite for lasing. Without population inversion, incident photons are absorbed; with population inversion, stimulated emission dominates over absorption, resulting in optical gain (amplification).

4.1.3 1.3 Derivation of Lasing Threshold Gain Condition

Consider a semiconductor laser with an active cavity of length L, bounded by two cleaved facets serving as partially reflecting mirrors with power reflectivities R_1 and R_2.

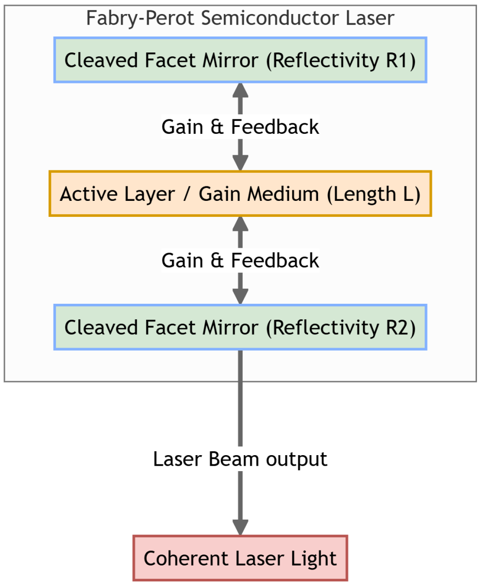

Figure 4.1: Fabry-Perot Laser Cavity

1. **Propagation Gain and Loss:** Let g be the optical gain coefficient per unit length and α_t represent the total internal absorption and scattering loss coefficient per unit length. As an optical wave propagates in the z-direction, its intensity grows according to:

$$I(z) = I(0)e^{(g-\alpha_t)z}$$

1. **Round-Trip Propagation:** In a complete round-trip through the cavity, the wave travels a distance of $2L$ and undergoes reflections from both facets. The intensity after one round-trip is:

$$I(2L) = I(0)R_1R_2e^{2(g-\alpha_t)L}$$

1. **Lasing Threshold:** Lasing occurs when the round-trip optical gain exactly compensates for the round-trip cavity losses, allowing the wave to maintain its intensity:

$$I(2L) = I(0) \implies R_1R_2e^{2(g_{th}-\alpha_t)L} = 1$$

(where g_{th} is the threshold gain).

1. Taking the natural logarithm on both sides:

$$2(g_{th} - \alpha_t)L = \ln\left(\frac{1}{R_1R_2}\right)$$

$$g_{th} - \alpha_t = \frac{1}{2L}\ln\left(\frac{1}{R_1R_2}\right)$$

1. Rearranging to solve for threshold gain g_{th}:

$$g_{th} = \alpha_t + \frac{1}{2L}\ln\left(\frac{1}{R_1R_2}\right)$$

Conclusion: The threshold gain g_{th} must equal the sum of the internal medium losses (α_t) and the mirror mirror-facet transmission losses ($\frac{1}{2L}\ln(1/R_1R_2)$).

4.2 2. Optical Detectors (PIN vs. APD)

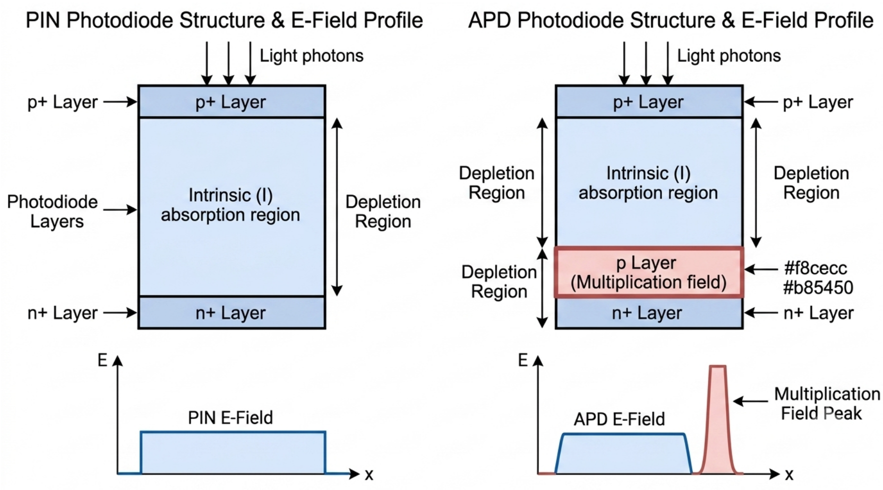

Figure 4.2: PIN vs. APD Structure Comparison

Photodetectors convert optical power into electrical current. The two main semiconductor detectors are PIN photodiodes and Avalanche Photodiodes (APDs).

4.2.1 2.1 Comparison Table

Feature / Criteria	PIN Photodiode	Avalanche Photodiode (APD)
Structure	P-type and N-type layers separated by a wide intrinsic (I) region.	P-I-N structure with an additional high-field multiplication region.
Internal Gain (M)	None ($M = 1$).	High internal gain ($M \approx 50$ to 100).
Responsivity (R)	Moderate (0.6 to 0.9 A/W).	High (10 to 100 A/W due to gain M).
Reverse Bias Voltage	Low (5 to 15 V).	High (100 to 300 V).
Receiver Sensitivity	Moderate.	High (10 to 15 dB higher than PIN).
Noise Profile	Low; limited by thermal noise of external circuits.	High; exhibits excess avalanche multiplication noise.
Typical Applications	Short/medium-reach links, LANs, receiver cards.	Long-haul transmission, undersea links, high-speed Rx.

4.2.2 2.2 APD Impact Ionization and Multiplication Mechanism

An Avalanche Photodiode provides internal gain through carrier multiplication in a high electric field region.

- **Mechanism:** The APD is operated under a high reverse bias voltage, creating an extremely strong electric field ($> 10^5$ V/cm) across a dedicated multiplication region.

- **Impact Ionization:** When an incident photon generates a primary electron-hole pair in the absorption region, the electron is accelerated by the high electric field. As it traverses the multiplication region, it collides with lattice atoms. If the electron has sufficient kinetic energy, it excites an electron from the valence band to the conduction band, creating a secondary electron-hole pair.
- **Avalanche Effect:** The newly generated carriers are also accelerated by the electric field and initiate further impact ionization events, creating a cascade (avalanche) process.
- **Gain Factor (M):** The total multiplication factor is defined as:

$$M = \frac{I_{out}}{I_{primary}}$$

4.2.3 2.3 Detector Material Selection for the 1550 nm Window

The absorption of light in a photodetector is governed by the semiconductor bandgap energy E_g. A photon is absorbed only if its energy exceeds the bandgap: $h\nu \geq E_g \implies \frac{hc}{\lambda} \geq E_g$. The **cutoff wavelength λ_c** is:

$$\lambda_c = \frac{hc}{E_g} \approx \frac{1.24}{E_g(\text{eV})}\ \mu\text{m}$$

- **Silicon (Si) Unsuitability:** Silicon has a bandgap energy $E_g \approx 1.1$ eV, corresponding to a cutoff wavelength $\lambda_c \approx 1.1\ \mu$m. Since photons in the 1550 nm transmission window ($\lambda = 1.55\ \mu$m, energy $E \approx 0.8$ eV) do not have enough energy to excite electrons across Silicon's bandgap, Si is transparent to this wavelength and cannot detect it.
- **Preferred Materials: Indium Gallium Arsenide (InGaAs)** ($E_g \approx 0.73$ eV, $\lambda_c \approx 1.7\ \mu$m) and **Germanium (Ge)** ($E_g \approx 0.67$ eV, $\lambda_c \approx 1.85\ \mu$m) are preferred. InGaAs is highly favored for long-haul networks because of its high quantum efficiency ($> 80\%$) and low dark current.

4.3 3. Detector Responsivity & Quantum Efficiency

4.3.1 3.1 Quantum Efficiency (η)

Quantum efficiency is the fraction of incident photons that generate electron-hole pairs collected at the detector terminals:

$$\eta = \frac{\text{Electron generation rate}}{\text{Photon incident rate}} = \frac{I_p/e}{P_{in}/h\nu} = \frac{I_p h\nu}{eP_{in}}$$

Where:

- I_p is the photogenerated current (photocurrent).
- P_{in} is the incident optical power.
- e is the electron charge (1.602×10^{-19} C).
- $h\nu$ is the energy of a single photon.

4.3.2 3.2 Responsivity (R)

Responsivity is the ratio of output photocurrent to incident optical power, representing the conversion efficiency:

$$R = \frac{I_p}{P_{in}}$$

Using the formula for quantum efficiency:

$$\eta = \frac{I_p h\nu}{eP_{in}} \implies I_p = \frac{\eta e P_{in}}{h\nu}$$

Substitute I_p into the responsivity definition:

$$R = \frac{\eta e}{h\nu} = \frac{\eta e \lambda}{hc}$$

Substituting values for e, h, and c, and expressing λ in micrometers (μm):

$$R \approx \frac{\eta \lambda(\mu\text{m})}{1.24} \text{ A/W}$$

4.4 4. Optical Receivers & Noise Sources

4.4.1 4.1 Digital Receiver Block Diagram

A digital optical receiver converts optical pulses back into electrical data:

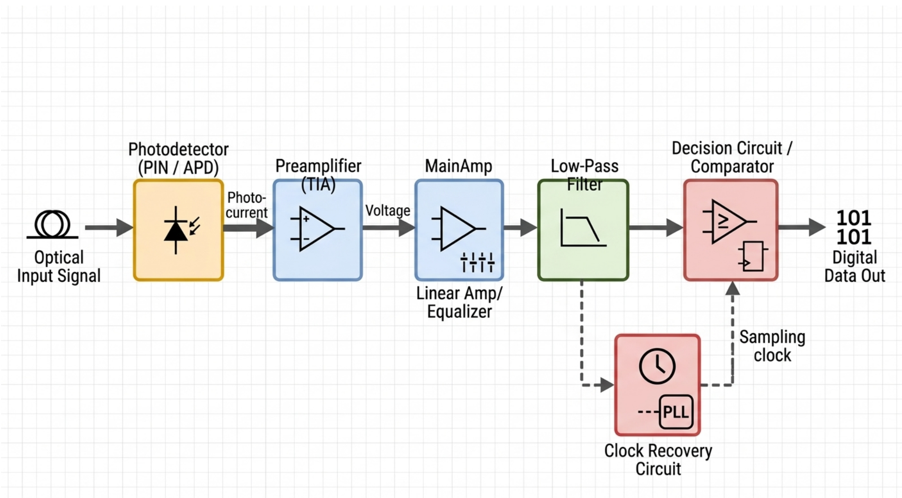

Figure 4.3: Digital Optical Receiver

1. **Photodetector:** PIN or APD converts the incident optical power waveform into a photocurrent.
2. **Preamplifier:** Amplifies the weak photocurrent, converting it to a voltage while minimizing noise (usually transimpedance design).
3. **Linear Channel (Equalizer & Main Amp):** Amplifies the signal and shapes the pulses (using an equalizer) to reduce inter-symbol interference.

4. **Filter:** Restricts the bandwidth of the signal to reject out-of-band noise.
5. **Decision Circuit:** A comparator samples the signal at clock intervals (reconstructed by a clock recovery circuit) to determine if a ”1” or ”0” was transmitted.

4.4.2 4.2 Noise Sources

The performance of a receiver is limited by noise, which fluctuates the electrical signal level at the decision circuit.

A. Shot Noise

Shot noise is a fundamental, signal-dependent noise caused by the discrete, random arrival of photons and generation of carrier pairs. The mean-square shot noise current is:

$$\langle i_{shot}^2 \rangle = 2e(I_p + I_d)M^2F(M)B$$

Where:

- I_p is the signal photocurrent.
- I_d is the dark current (current flowing when no light is incident).
- M is the APD gain (for PIN, $M = 1, F(M) = 1$).
- $F(M)$ is the excess noise factor ($F(M) \approx M^x$, where $0 < x \leq 1$).
- B is the receiver bandwidth.

B. Thermal Noise (Johnson Noise)

Thermal noise is caused by the random thermal motion of electrons in the receiver load resistor R_L:

$$\langle i_{thermal}^2 \rangle = \frac{4k_BTB}{R_L}$$

Where k_B is Boltzmann's constant and T is the absolute temperature.

C. Amplifier Noise

Noise generated by active transistors in the preamplifier stage, represented by a noise figure.

4.5 5. Optical Link Design

4.5.1 5.1 Link Power Budget

The **Link Power Budget** ensures that the optical transmitter launches sufficient power to overcome all link losses and deliver adequate power to the receiver.

- **Design Equation:**

$$P_T - P_R = \alpha L + \alpha_{sp}N_{sp} + \alpha_c N_c + M_s$$

Where:

- P_T is the transmitter launched power (dBm).
- P_R is the receiver sensitivity required to maintain a target BER (dBm).
- α is the fiber attenuation coefficient (dB/km).
- L is the fiber length (km).
- α_{sp} and N_{sp} are the loss per splice (dB) and number of splices.
- α_c and N_c are the loss per connector (dB) and number of connectors.
- M_s is the system safety margin (typically 3 to 6 dB, accounting for component aging).

4.5.2 5.2 Rise-Time Budget

The **Rise-Time Budget** ensures the system is fast enough to support the target transmission speed, preventing dispersion-induced pulse overlap.

- **Design Equation:**

$$t_{sys} = \left(t_{tx}^2 + t_{mat}^2 + t_{mod}^2 + t_{rx}^2\right)^{1/2}$$

Where:

- t_{tx} is the transmitter rise-time (determined by source electronics).
- t_{mat} is the material dispersion rise-time: $t_{mat} = D_M \cdot L \cdot \Delta\lambda$.
- t_{mod} is the modal dispersion rise-time: $t_{mod} = \Delta t_{modal}$.
- t_{rx} is the receiver rise-time: $t_{rx} = 350/B_{rx}$ ps (where B_{rx} is the receiver electrical bandwidth in MHz).
- **System Limits:**
- For NRZ coding: $t_{sys} \leq 0.7T = \frac{0.7}{B_T}$
- For RZ coding: $t_{sys} \leq 0.35T = \frac{0.35}{B_T}$

(where T is the bit period and B_T is the bit rate).

4.5.3 5.3 Quantum Limit of Detection

The quantum limit is the minimum number of photons required per pulse to achieve a target Bit Error Rate (BER) in an ideal receiver (zero thermal noise, zero dark current, and 100% quantum efficiency).

1. **Poisson Distribution:** The probability of generating n electron-hole pairs from a pulse containing an average of N_p photons is given by the Poisson distribution:

$$P(n) = \frac{N_p^n e^{-N_p}}{n!}$$

1. **Bit Errors:** In digital transmission, we assume a "0" bit contains no photons. An error occurs only when a "1" bit is transmitted, but it generates 0 electron-hole pairs, causing the receiver to register a "0" (a miss).

2. The probability of error P_e (detecting 0 electrons when a "1" is sent) is:

$$P_e = P(0) = \frac{N_p^0 e^{-N_p}}{0!} = e^{-N_p}$$

1. For a target BER of 10^{-9}:

$$e^{-N_p} = 10^{-9} \implies -N_p = \ln(10^{-9}) \implies N_p \approx 20.7 \approx 21 \text{ photons}$$

Conclusion: An ideal receiver requires an average of at least **21 photons** per "1" pulse to achieve a Bit Error Rate of 10^{-9}.

4.6 6. Worked Practice Problems

4.6.1 Problem 6.1: Quantum Efficiency & Responsivity

When 3×10^{11} photons of wavelength 0.85 μm are incident on a photodiode, an average of 1.2×10^{11} electrons are collected at the terminals. Calculate:

1. The quantum efficiency (η) of the photodiode.

2. The responsivity (R) of the photodiode at 0.85 μm.

Solution: 1. Given Data:

- Number of incident photons = 3×10^{11}
- Number of collected electrons = 1.2×10^{11}
- Wavelength (λ) = 0.85 μm

2. Calculate Quantum Efficiency (η):

$$\eta = \frac{\text{Collected electrons}}{\text{Incident photons}} = \frac{1.2 \times 10^{11}}{3 \times 10^{11}} = 0.40 = 40\%$$

3. Formula for Responsivity (R):

$$R = \frac{\eta\lambda(\mu\text{m})}{1.24}$$

4. Calculation:

$$R = \frac{0.40 \times 0.85}{1.24} = \frac{0.34}{1.24} \approx 0.274 \text{ A/W}$$

Final Answer:

1. The quantum efficiency is 40%.

2. The responsivity is 0.274 **A/W**.

4.6.2 Problem 6.2: Link Power Budget

An optical fiber link is 50 km long. The optical source launches a power of $P_T = -3$ dBm into the fiber. The fiber attenuation is $\alpha = 0.25$ dB/km. The link includes 5 splices with a loss of 0.1 dB each, and 2 connectors with a loss of 0.5 dB each. If the system safety margin is 4 dB, calculate the minimum receiver sensitivity (P_R) required for the link to operate.

Solution: 1. Given Data:

- $L = 50$ km
- $P_T = -3$ dBm
- $\alpha = 0.25$ dB/km
- $N_{sp} = 5$, $\alpha_{sp} = 0.1$ dB
- $N_c = 2$, $\alpha_c = 0.5$ dB
- $M_s = 4$ dB

2. Formula:

$$P_T - P_R = \alpha L + N_{sp}\alpha_{sp} + N_c\alpha_c + M_s$$

3. Calculate Total Link Loss:

$$\text{Total Loss} = (0.25 \times 50) + (5 \times 0.1) + (2 \times 0.5) + 4$$

$$\text{Total Loss} = 12.5 + 0.5 + 1.0 + 4 = 18.0 \text{ dB}$$

4. Calculate Receiver Sensitivity (P_R):

$$-3 \text{ dBm} - P_R = 18.0 \text{ dB}$$

$$-P_R = 18.0 + 3 \implies P_R = -21.0 \text{ dBm}$$

Final Answer: The minimum receiver sensitivity required is -21.0 **dBm** (any receiver with sensitivity ≤ -21 dBm will work).

4.7 7. Laser Cavity Physics, Materials, and Worked Problems (Q4.7) [10M][]

Understanding the material science, junction physics, and resonator dynamics of semiconductor light sources is key to optimizing optical transmitters.

4.7.1 7.1 Materials and Junction Physics

A. Direct vs. Indirect Band-Gap Semiconductors

- **Direct Band-Gap (e.g., GaAs, InP):** The conduction band minimum and valence band maximum occur at the same value of electron momentum (crystal momentum vector $\vec{k}$). When an electron recombines with a hole, it can drop directly across the bandgap, releasing its energy almost entirely as a **photon**. The recombination rate is extremely high (transition lifetime of nanoseconds), making these materials ideal for light sources.

- **Indirect Band-Gap (e.g., Silicon, Germanium):** The conduction band minimum and valence band maximum occur at different values of electron momentum. Recombination requires a simultaneous change in momentum, which must be assisted by a lattice vibration (absorbing or emitting a phonon). This three-body interaction is highly inefficient and slow (transition lifetime of milliseconds), with energy released as **heat** rather than light. Hence, they are poor optical sources.

Note on Intrinsic Semiconductors:* A perfect semiconductor crystal containing no impurities or lattice defects is called an **intrinsic semiconductor**. When light emission results from applying an electric field, the process is known as **electroluminescence**.

B. Heterojunctions: Isotype and Anisotype

A heterojunction is an interface between two different semiconductor materials with different bandgap energies. There are two primary categories:

1. **Isotype Heterojunctions:** An interface between two semiconductors of the **same conductivity type** but different bandgaps (e.g., n-type GaAs and N-type AlGaAs, denoted as n-N, or p-P).

2. **Anisotype Heterojunctions:** An interface between two semiconductors of **different conductivity types** and different bandgaps (e.g., p-type GaAs and N-type AlGaAs, denoted as p-N, or n-P).

- **Significance:** Double heterostructures (DH) use combinations of isotype and anisotype junctions to create potential energy wells that confine both injected carriers (electrons and holes) and the optical field to a thin active layer, reducing the threshold current density. The **GaAs/AlGaAs DH** system is the most mature for fabricating lasers and LEDs in the shorter wavelength region (0.8 to 0.9 μm).

C. Doping for Population Inversion

To achieve population inversion at a p-n junction, **heavy doping of both the n-type and p-type regions** is required (producing degenerate semiconductors). When heavily forward-biased, the Fermi levels are pushed deep into the conduction and valence bands, injecting a massive concentration of electrons and holes into the junction's depletion layer, creating the active population inversion layer.

4.7.2 7.2 Worked Numerical Problems

Problem 1: Cavity Facet Reflectivity

A GaAs injection laser with an optical cavity has a refractive index of $n = 3.6$. Calculate the reflectivity for normal incidence of the plane wave on the GaAs-air interface.

Solution:

1. **Formula for Reflectivity** (R):

$$R = \left(\frac{n-1}{n+1}\right)^2$$

1. **Calculation:**

$$R = \left(\frac{3.6-1}{3.6+1}\right)^2 = \left(\frac{2.6}{4.6}\right)^2 \approx (0.5652)^2 \approx \mathbf{0.32} \quad (32\%)$$

Final Answer: The reflectivity at the GaAs-air interface is 0.32 (or 32%).

Problem 2: Longitudinal Modes in Ruby Laser

A ruby laser has a crystal of length 3 cm with a refractive index of 1.60 and a wavelength of 0.43 μm. Determine the number of longitudinal modes.

Solution:

1. **Given Data:**

- $L = 3$ cm $= 0.03$ m
- $n = 1.60$
- $\lambda = 0.43\ \mu\text{m} = 0.43 \times 10^{-6}$ m

1. **Formula for Mode Number (m):**

$$m = \frac{2nL}{\lambda}$$

1. **Calculation:**

$$m = \frac{2 \times 1.60 \times 0.03}{0.43 \times 10^{-6}} = \frac{0.096}{0.43 \times 10^{-6}} \approx \mathbf{2.23 \times 10^5}$$

Final Answer: The number of longitudinal modes is 2.23×10^5 (Option c).

Problem 3: Laser Mode Frequency Separation

A semiconductor laser crystal of length 5 cm and refractive index 1.8 is used as an optical source. Determine the frequency separation of the modes.

Solution:

1. **Given Data:**

- $L = 5$ cm $= 0.05$ m
- $n = 1.8$
- $c \approx 3 \times 10^8$ m/s

1. **Formula for Frequency Separation (Δf):**

$$\Delta f = \frac{c}{2nL}$$

1. **Calculation:**

$$\Delta f = \frac{3 \times 10^8 \text{ m/s}}{2 \times 1.8 \times 0.05 \text{ m}} = \frac{3 \times 10^8}{0.18} \approx 1.67 \times 10^9 \text{ Hz} = \mathbf{1.67} \text{ GHz}$$

Final Answer: The frequency separation of the modes is 1.67 **GHz** (closest option in CA4 is 1.6 **GHz**).

Problem 4: Wavelength Mode Separation

Calculate the mode separation in terms of free-space wavelength for a laser operating at $\lambda = 0.5\ \mu$m if its mode frequency separation is $\Delta f = 2$ GHz.

Solution:

1. **Given Data:**

- $\lambda = 0.5\ \mu\text{m} = 5 \times 10^{-7}$ m
- $\Delta f = 2\ \text{GHz} = 2 \times 10^{9}$ Hz
- $c \approx 3 \times 10^{8}$ m/s

1. **Formula for Wavelength Separation ($\Delta\lambda$):**

$$\Delta\lambda = \frac{\lambda^2 \Delta f}{c}$$

1. **Calculation:**

$$\Delta\lambda = \frac{(5 \times 10^{-7}\ \text{m})^2 \times 2 \times 10^{9}\ \text{s}^{-1}}{3 \times 10^{8}\ \text{m/s}} = \frac{25 \times 10^{-14} \times 2 \times 10^{9}}{3 \times 10^{8}} = \frac{50 \times 10^{-5}}{3 \times 10^{8}} \approx \mathbf{1.67 \times 10^{-12}}\ \text{m}$$

Final Answer: The mode separation in terms of free-space wavelength is 1.67×10^{-12} **m** (or 1.67 **pm**; closest option in CA4 is 1.6×10^{-12} **m**).

Problem 5: Radiative Minority Carrier Lifetime

Calculate the radiative minority carrier lifetime in gallium arsenide when the minority carriers are electrons injected into a p-type region which has a hole concentration of $p_0 = 10^{18}\ \text{cm}^{-3}$. The recombination coefficient for gallium arsenide is $B = 7.21 \times 10^{-10}\ \text{cm}^3\text{s}^{-1}$.

Solution:

1. **Given Data:**

- $p_0 = 10^{18}\ \text{cm}^{-3}$
- $B = 7.21 \times 10^{-10}\ \text{cm}^3\text{s}^{-1}$

1. **Formula for Radiative Lifetime (τ_r):**

$$\tau_r = \frac{1}{B \cdot p_0}$$

1. **Calculation:**

$$\tau_r = \frac{1}{(7.21 \times 10^{-10}\ \text{cm}^3\text{s}^{-1}) \times (10^{18}\ \text{cm}^{-3})} = \frac{1}{7.21 \times 10^{8}\ \text{s}^{-1}} \approx 1.387 \times 10^{-9}\ \text{s} \approx \mathbf{1.39}\ \text{ns}$$

Final Answer: The radiative minority carrier lifetime is 1.39 **ns** (Option c).

Chapter 5

Optical Switches & Amplifiers

This chapter compiles study materials and detailed exam answers for **Unit V: Optical Switches & Amplifiers** (including WDM systems and networks). It covers optical amplifiers (EDFA and Raman), Coupled Mode Theory in directional couplers, Lithium Niobate electro-optic switches, WDM/DWDM systems, WDM network elements (OADM, OXC, RWA, wavelength conversion), and SONET transport layouts.

5.1 1. Optical Amplifiers (EDFA vs. Raman Amplifiers)

Optical amplifiers amplify optical signals directly in the optical domain without electronic regeneration (optical-to-electrical-to-optical conversion). The two most important optical amplifiers are Erbium-Doped Fiber Amplifiers (EDFAs) and Raman Amplifiers.

5.1.1 1.1 Comparison Table

Feature / Criteria	Erbium-Doped Fiber Amplifier (EDFA)	Raman Amplifier
Gain Medium	Silica fiber doped with Erbium (Er^{3+}) ions (lumped gain).	Standard transmission silica fiber (distributed gain).
Amplification Mechanism	Stimulated emission of excited Erbium ions.	Stimulated Raman Scattering (SRS) (pump photons transfer energy to signal photons).
Gain Band	Fixed; C-band (1530 to 1565 nm) and L-band (1565 to 1625 nm).	Highly flexible; can operate at any wavelength depending on the pump.
Typical Pump Wavelengths	980 nm or 1480 nm.	Must be $\approx$ 100 nm shorter than signal (Stokes shift $\approx$ 13.2 THz).
Noise Figure	Low (3 to 5 dB).	Extremely low (effectively negative or near-zero due to distributed gain).
Pump Power Requirement	Moderate (10 to 100 mW).	Very high (100 mW to several Watts).
Cost & Complexity	Lower cost, simple design.	Higher cost, complex high-power pump lasers and controllers.

5.1.2 1.2 Erbium-Doped Fiber Amplifier (EDFA)

EDFAs operate by exploiting the three-level energy transition system of Erbium (Er^{3+}) ions embedded in a silica fiber core.

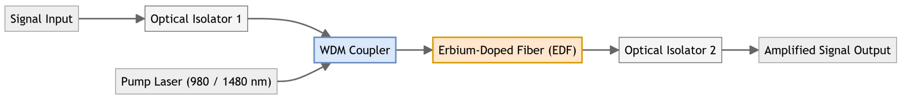

Figure 5.1: EDFA Block Diagram

- **Pumping at** 980 **nm:** Er^{3+} ions are excited from the ground state ($^4I_{15/2}$) to the short-lived excited state ($^4I_{11/2}$). From there, they undergo rapid, non-radiative decay ($\approx 1\ \mu$s) to the metastable state ($^4I_{13/2}$).
- **Pumping at** 1480 **nm:** Er^{3+} ions are excited directly from the ground state to the metastable state ($^4I_{13/2}$).
- **Stimulated Emission:** The metastable state has a long lifetime ($\approx$ 10 ms). This long lifetime allows a population inversion to build up between the metastable state and the ground state. When signal photons (in the 1550 nm band) pass through the fiber, they trigger stimulated transition of the Erbium ions from the metastable state back to the ground state, emitting coherent signal photons and providing optical gain.

5.1.3 1.3 Raman Amplifiers

- **Mechanism:** Based on **Stimulated Raman Scattering (SRS)**, a non-linear process where an intense pump wave launched into the fiber interacts with the silica glass vibrational modes (optical phonons).
- **Energy Transfer:** A pump photon of frequency ν_p is scattered by a silica molecule. The molecule absorbs a portion of the photon's energy as molecular vibrational energy (phonon of frequency ν_{vib}), and the remaining energy is emitted as a lower-energy scattered photon of frequency $\nu_s = \nu_p - \nu_{vib}$ (Stokes frequency).
- **Distributed Amplification:** If a signal co-propagates at the Stokes frequency, the transfer of energy from the pump to the signal occurs continuously along the transmission fiber, providing distributed optical gain.

5.2 2. Directional Couplers & Coupled Mode Analysis

A 2×2 directional coupler consists of two parallel optical waveguides fabricated close to each other, allowing power exchange through evanescent wave coupling.

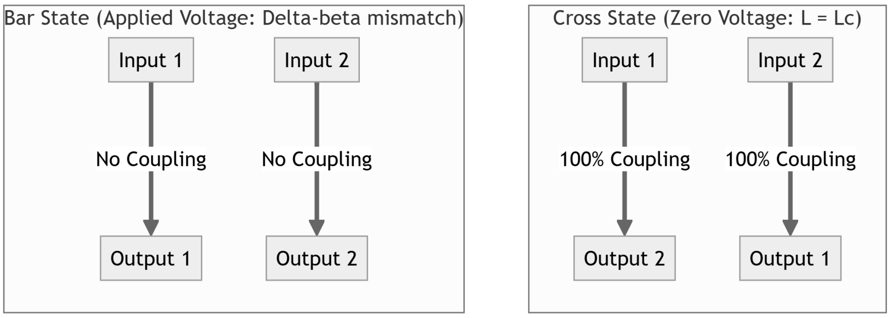

Figure 5.2: Directional Coupler Switch States

5.2.1 2.1 Coupled Mode Formulation

Let $a_1(z)$ and $a_2(z)$ be the mode amplitudes in Waveguide 1 and Waveguide 2, respectively. The coupled differential equations governing the propagation of mode amplitudes along the propagation direction z are:

$$\frac{da_1}{dz} = -j\beta_1 a_1 - j\kappa a_2$$
$$\frac{da_2}{dz} = -j\beta_2 a_2 - j\kappa a_1$$

Where:

- β_1, β_2 are the propagation constants of Waveguides 1 and 2.
- κ is the coupling coefficient between the two guides (determined by waveguide geometry, refractive index profile, and waveguide separation).

5.2.2 2.2 Derivation of Power Transfer for Phase-Matched Guides ($\beta_1 = \beta_2 = \beta$)

1. Assume identical waveguides, so the phase mismatch is zero ($\Delta\beta = \beta_1 - \beta_2 = 0$).
2. The coupled equations become:

$$\frac{da_1}{dz} = -j\beta a_1 - j\kappa a_2 \quad \text{— (Eq 5.1)}$$
$$\frac{da_2}{dz} = -j\beta a_2 - j\kappa a_1 \quad \text{— (Eq 5.2)}$$

1. Assume power P_0 is launched into Waveguide 1 at $z = 0$:

$$a_1(0) = \sqrt{P_0}, \quad a_2(0) = 0$$

1. Differentiating Eq 5.1 and substituting Eq 5.2:

$$\frac{d^2 a_1}{dz^2} + 2j\beta \frac{da_1}{dz} + (\beta^2 - \kappa^2) a_1 = 0$$

1. Solving this second-order differential equation with the boundary conditions yields:

$$a_1(z) = \sqrt{P_0}\cos(\kappa z)e^{-j\beta z}$$
$$a_2(z) = -j\sqrt{P_0}\sin(\kappa z)e^{-j\beta z}$$

1. The optical power in each waveguide is:

$$P_1(z) = |a_1(z)|^2 = P_0\cos^2(\kappa z)$$
$$P_2(z) = |a_2(z)|^2 = P_0\sin^2(\kappa z)$$

5.2.3 2.3 Coupling Length and Switching States

- **Coupling Length (L_c):** The length required for 100% power transfer from Guide 1 to Guide 2:

$$\kappa L_c = \frac{\pi}{2} \implies L_c = \frac{\pi}{2\kappa}$$

- **Cross State (ON):** The physical length of the coupler is chosen to be $L = L_c$. Light launched into Port 1 transfers completely to Port 2 (cross path).

- **Bar State (OFF):** An external voltage is applied across electrodes. Through the electro-optic effect, the refractive index of one waveguide increases while the other decreases, creating a phase mismatch ($\Delta\beta = \beta_1 - \beta_2 \neq 0$). Under this condition, the power coupled into Guide 2 is:

$$P_2(z) = P_0\frac{\kappa^2}{\kappa^2 + (\Delta\beta/2)^2}\sin^2\left(\sqrt{\kappa^2 + (\Delta\beta/2)^2}z\right)$$

By selecting the voltage such that $\sqrt{\kappa^2 + (\Delta\beta/2)^2}L = \pi$, the power coupled to Guide 2 drops to zero ($P_2(L) = 0$), forcing all light to exit through the launch guide Port 1 (bar path).

5.3 3. Electro-Optic Switches (Lithium Niobate MZI)

High-speed optical switches are fabricated on Lithium Niobate ($LiNbO_3$) substrates because of the material's strong linear electro-optic (Pockels) effect.

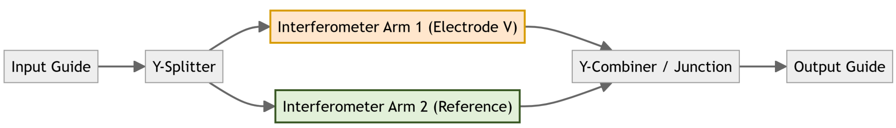

Figure 5.3: MZI Switch Layout

- **Structure:** A Mach-Zehnder Interferometer (MZI) consists of an input Y-splitter that divides the optical power equally into two parallel waveguide arms. The arms run parallel over a length L before recombining at an output Y-junction. Electrodes are deposited along the arms.

- **Working Principle:**

- **Zero Voltage (Constructive Interference):** Light travels at the same speed in both arms, arriving in-phase at the output junction. The waves interfere constructively, exiting the device (ON state).

- **Applied Voltage (Destructive Interference):** A voltage is applied across the electrodes. The electric field induces a refractive index change $\Delta n \propto E$, which creates a relative phase difference $\Delta\phi$ between the arms:

$$\Delta\phi = \frac{2\pi\Delta nL}{\lambda}$$

By applying the **switching voltage** (V_{π}) that induces a phase difference of $\Delta\phi = \pi$, the waves interfere destructively at the output Y-junction, radiating the optical power into the substrate (OFF state).

5.4 4. WDM & DWDM Systems

Wavelength Division Multiplexing (WDM) combines multiple optical carrier signals of different wavelengths onto a single optical fiber, increasing the link's overall capacity.

5.4.1 4.1 Comparison Table

Feature / Criteria	Coarse WDM (CWDM)	Dense WDM (DWDM)
Channel Spacing	Wide; typically 20 nm.	Narrow; $\leq$ 1.6 nm (100 GHz or 50 GHz grids).
Number of Channels	Low; typically 8 to 16 channels.	High; 80 to 160+ channels.
Laser Requirements	Uncooled, low-precision, inexpensive lasers.	Temperature-stabilized, high-precision DFB lasers.
Optical Filters	Wide bandpass filters, low cost.	Narrow bandpass filters, high performance.
Typical Range	Short-reach (metro access, LANs, $<$ 50 km).	Long-haul, regional networks ($>$ 100 km).

5.4.2 4.2 WDM Components

- **Multiplexer (MUX):** Combines multiple input wavelengths onto a single output fiber.
- **Demultiplexer (DEMUX):** Separates the multiplexed wavelengths from a single input fiber onto separate output fibers.
- **Fused Biconical Taper (FBT) Couplers:** Fabricated by fusing and stretching two parallel fibers. They act as splitters/combiners, separating or coupling power through evanescent wave overlap.

5.4.3 4.3 Unidirectional vs. Bidirectional WDM

- **Unidirectional WDM:** Wavelengths propagate in a single direction along a fiber link. Two fibers are required for duplex communication (one fiber for transmit, one fiber for receive).

- **Bidirectional WDM:** Wavelengths propagate in both directions along a single fiber (e.g., $\lambda_1, \lambda_2, \lambda_3$ transmit from West to East, and $\lambda_4, \lambda_5, \lambda_6$ transmit from East to West). This halves the required fiber count but requires optical filters to prevent crosstalk from backscattering.

5.5 5. Principles of WDM Networks

Wavelength-routed mesh networks operate by routing signals directly in the optical domain based on their wavelength.

5.5.1 5.1 Optical Add-Drop Multiplexers (OADM)

***Function:** Used at intermediate nodes to selectively* drop *(extract) one or more wavelength channels from a multiplexed fiber, and* add* new signals on those same wavelengths, while letting all other channels pass through transparently.

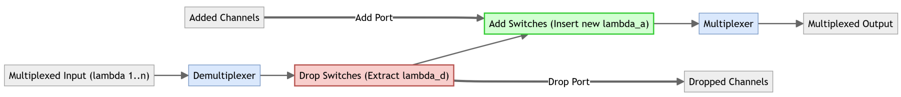

Figure 5.4: OADM Block Diagram

5.5.2 5.2 Optical Cross-Connects (OXC)

- **Function:** Operates at core network nodes to route optical signals from any input port/fiber to any output port/fiber on a wavelength-by-wavelength basis. It performs space-division and wavelength routing entirely in the optical domain.

Figure 5.5: OXC Wavelength Routing

5.5.3 5.3 Routing and Wavelength Assignment (RWA) Problem

RWA is the challenge of routing an optical connection (lightpath) from source to destination and assigning a specific wavelength to it.

- **Wavelength Continuity Constraint:** In a transparent optical network, a lightpath must use the same wavelength on all fiber links along its route. If no single wavelength is free along the entire path, the connection is blocked, even if individual links have free capacity.

5.5.4 5.4 Wavelength Conversion

- **Definition:** Translates the carrier wavelength of an optical signal from λ_1 to λ_2 at a network node without converting it to the electrical domain.
- **Significance:** It eliminates the wavelength continuity constraint. By allowing a lightpath to change wavelengths at intermediate nodes, wavelength conversion reduces blocking probability and increases network routing flexibility.

5.6 6. SONET Payload over WDM

SONET (Synchronous Optical Network) is a standard for synchronous data transmission over optical fibers. WDM networks transport SONET payloads transparently.

5.6.1 6.1 SONET Protocol Layers

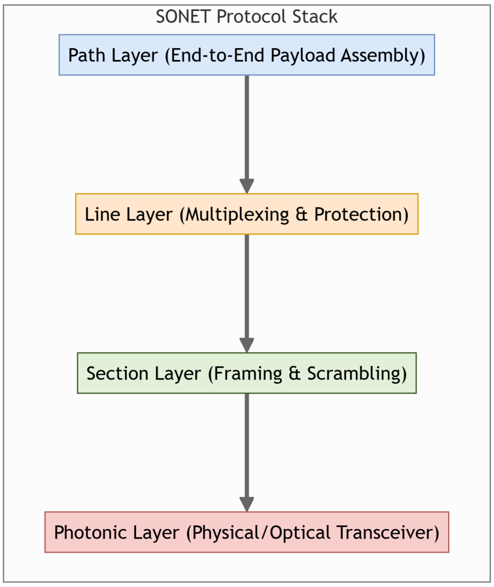

Figure 5.6: SONET Protocol Layer Stack

- **Photonic (Physical) Layer:** Manages physical transmission parameters, converting electrical signals into optical pulses.
- **Section Layer:** Manages transmission over a single physical fiber link, handling framing, scrambling, and basic link error monitoring.
- **Line Layer:** Manages transport between multiplexers, handling synchronization, protection switching, and payload multiplexing.
- **Path Layer:** Manages end-to-end transport between path terminating equipment (PTE), handling packet assembly, payload mapping, and end-to-end error checking.

Chapter 6

Nonlinear Effects

This chapter compiles study materials and detailed exam answers for **Unit VI: Nonlinear Effects** in optical fibers. It covers the Kerr effect (intensity-dependent refractive index), nonlinear scattering mechanisms (Stimulated Raman Scattering vs. Stimulated Brillouin Scattering), refractive-index-based nonlinearities (SPM, XPM, FWM), Group Velocity Dispersion (GVD), and the principles of optical soliton propagation in high-speed networks.

6.1 1. The Kerr Effect & Optical Refractive Index

The Kerr effect refers to the change in the refractive index of a material in response to an applied electric field, specifically showing a linear dependence on the optical intensity of the propagating lightwave (also known as the optical Kerr effect).

6.1.1 1.1 Mathematical Formulation

[5M][] Q6.1 (Kerr Effect Definition & Refractive Index Variation)

- **Definition:** In silica fibers, the refractive index n experienced by an optical signal is intensity-dependent and is expressed as:

$$n(I) = n_0 + n_2 I$$

Where:*

- n_0 is the linear refractive index of silica glass ($\approx$ 1.45 to 1.48).
- n_2 is the nonlinear refractive index coefficient of silica glass ($\approx 2.6 \times 10^{-20}$ m^2/W).
- I is the optical intensity in the fiber core, defined as:

$$I = \frac{P}{A_{\text{eff}}}$$

(with P representing the optical power and A_{eff} representing the effective mode-field area).

- **Expression in terms of Power:**

$$n(P) = n_0 + \bar{n}_2 P$$

Where:*

- $\bar{n}_2 = \frac{n_2}{A_{\text{eff}}}$ (in units of W^{-1}m^{-1}).

6.1.2 1.2 Nonlinear Phenomena Caused by the Kerr Effect

Because the refractive index fluctuates with power variations, several third-order susceptibility ($\chi^{(3)}$) nonlinear phenomena occur:

1. **Self-Phase Modulation (SPM):** A single pulse alters its own phase profile, causing spectral broadening.
2. **Cross-Phase Modulation (XPM):** The intensity fluctuations of one wavelength channel alter the phase of co-propagating channels, inducing inter-channel crosstalk.
3. **Four-Wave Mixing (FWM):** Three channels beat together to generate new intermodulation sidebands, severely degrading WDM networks.

6.2 2. Nonlinear Scattering (SRS vs. SBS)

Nonlinear scattering effects involve the inelastic scattering of pump photons to lower-frequency Stokes photons, accompanied by the excitation of acoustic or optical vibrational states in the silica lattice.

6.2.1 2.1 Comparison Table

[5M][] Q6.3 (a) (SRS vs. SBS Mechanism & Comparison)

Feature / Criteria	Stimulated Raman Scattering (SRS)	Stimulated Brillouin Scattering (SBS)
Physical Mechanism	Interaction of light with optical molecular vibrations (optical phonons).	Interaction of light with acoustic density waves/sound waves (acoustic phonons).
Phonon Energy	High energy ($\approx$ 100 meV), localized molecular vibrations.	Low energy ($\approx 50\ \mu$eV), traveling acoustic wave.
Scattering Direction	Occurs in both **forward** and **backward** directions.	Occurs **strictly in the backward** direction in single-mode fibers.
Frequency Shift (f_{shift})	Very large; $\approx$ 13.2 THz (Stokes shift in silica).	Very small; $\approx$ 10 to 11 GHz in silica.
Gain Bandwidth	Extremely broad ($\approx$ 10 to 20 THz).	Extremely narrow ($\approx$ 10 to 100 MHz).
Threshold Pump Power	Very high ($\approx$ 500 to 1000 mW).	Very low ($\approx$ 1 to 10 mW under CW illumination).

6.2.2 2.2 Threshold Power Approximations

The threshold pump power (P_{th}) for the onset of stimulated scattering is the power level at which the scattered Stokes power equals the pump power at the fiber output.

- **SBS Threshold Power Equation:**

$$P_{th,\text{SBS}} \approx \frac{21A_{\text{eff}}}{g_B L_{\text{eff}}}$$

Where:*

- g_B is the Brillouin gain coefficient ($\approx 5 \times 10^{-11}$ m/W).
- L_{eff} is the effective fiber interaction length:

$$L_{\text{eff}} = \frac{1 - e^{-\alpha L}}{\alpha}$$

- **SRS Threshold Power Equation:**

$$P_{th,\text{SRS}} \approx \frac{16 A_{\text{eff}}}{g_R L_{\text{eff}}}$$

Where:*

- g_R is the Raman gain coefficient ($\approx 1 \times 10^{-13}$ m/W).

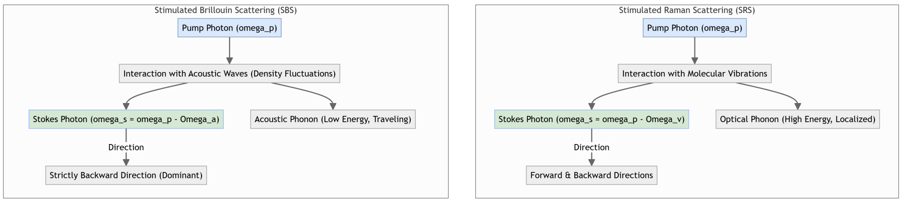

Figure 6.1: SRS vs. SBS Scattering Processes

6.2.3 2.3 Mitigation Strategies

1. **SBS Mitigation:**

- **Phase Modulation / Line Broadening:** Modulating the optical source phase broadens the laser linewidth ($\Delta\nu$), which reduces the effective Brillouin gain since gain $\propto \frac{1}{\Delta\nu_s + \Delta\nu_p}$.
- **Keep Power Below Threshold:** Restricting the input channel power below the 10 mW threshold.

1. **SRS Mitigation:**

- **Channel Power Control:** Ensuring individual channel powers in WDM systems remain below the Raman threshold ($P < 500$ mW).
- **Wavelength Band Separation:** Restricting WDM total band allocation width to minimize energy transfer from lower-wavelength channels to higher-wavelength channels.

6.3 3. Refractive-Index-Based Nonlinearities (SPM, XPM, FWM)

Refractive-index-based nonlinearities do not involve energy transfer to the fiber medium; instead, they originate from the intensity-dependent phase modulation of lightwaves.

6.3.1 3.1 Self-Phase Modulation (SPM)

[15M][] Q6.3 (b) (Self-Phase Modulation Mechanism & Phase Shift)

- **Physical Mechanism:** When an optical pulse propagates through a fiber, the local optical intensity $I(t)$ varies across the pulse envelope. Due to the Kerr effect, this creates a time-varying refractive index profile $n(t) = n_0 + n_2 I(t)$.

- **Nonlinear Phase Shift:** The propagation constant becomes time-dependent, generating a nonlinear phase shift $\phi_{\mathrm{NL}}(t)$ over a transmission length z:

$$\phi_{\mathrm{NL}}(t) = \gamma P(t) z$$

Where:*

- $\gamma = \frac{2\pi n_2}{\lambda A_{\mathrm{eff}}}$ is the fiber nonlinearity parameter (in $\mathrm{W}^{-1}\mathrm{km}^{-1}$).
- $P(t)$ is the time-dependent pulse power.

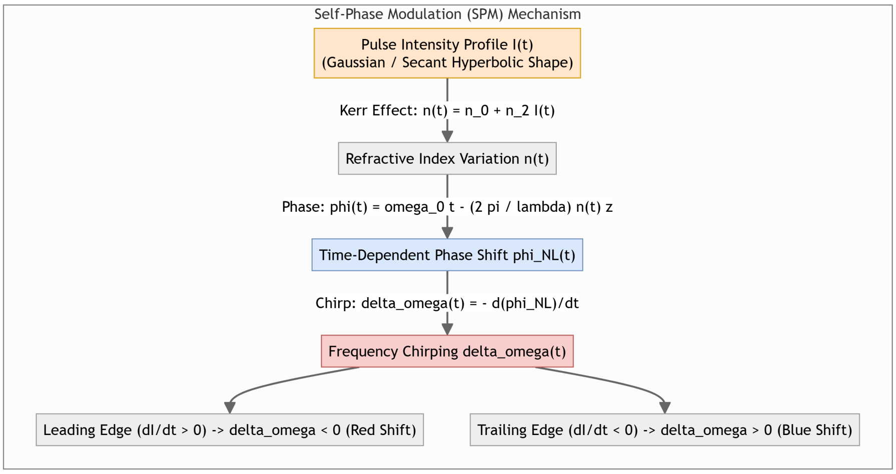

Figure 6.2: Self-Phase Modulation Chirping

- **Frequency Chirping:** The time-varying phase shift creates a frequency shift across the pulse, known as frequency chirp $\delta\omega(t)$:

$$\delta\omega(t) = -\frac{\partial \phi_{\mathrm{NL}}(t)}{\partial t} = -\gamma z \frac{\partial P(t)}{\partial t}$$

- **Leading Edge ($\frac{\partial P}{\partial t} > 0$): $\delta\omega < 0$**, which corresponds to a shift toward lower frequencies (**Red Shift**).
- **Trailing Edge ($\frac{\partial P}{\partial t} < 0$): $\delta\omega > 0$**, which corresponds to a shift toward higher frequencies (**Blue Shift**).
- **Spectral Broadening:** As the pulse propagates, new frequencies are continuously generated at the leading and trailing edges, leading to broadening of the pulse spectrum while the temporal width remains unchanged (in the absence of dispersion).

6.3.2 3.2 Cross-Phase Modulation (XPM)

- **Definition:** In multi-channel WDM systems, the intensity fluctuations of one wavelength channel (λ_1) alter the refractive index of the fiber core, which in turn modulates the phase of another co-propagating channel (λ_2).

- **Equation:** The total nonlinear phase shift for channel i in a multi-channel system is:

$$\phi_{\mathrm{NL},i} = \gamma L_{\mathrm{eff}} \left(P_i + 2 \sum_{j \neq i} P_j \right)$$

Note:* The factor of 2 indicates that XPM is twice as effective as SPM at inducing phase shifts.

- **Impact:** XPM induces amplitude jitter and spectral broadening, causing severe inter-channel crosstalk.

6.3.3 3.3 Four-Wave Mixing (FWM)

- **Definition:** A third-order parametric nonlinear process where three optical waves of frequencies f_1, f_2, and f_3 interact in a fiber medium to generate a fourth wave of frequency $f_{ijk} = f_i + f_j - f_k$.

- **Phase-Matching Condition:** For FWM to occur efficiently, the propagation constants must satisfy:

$$\Delta k = k_i + k_j - k_k - k_{ijk} \approx 0$$

This condition is easily met near the zero-dispersion wavelength of the fiber.

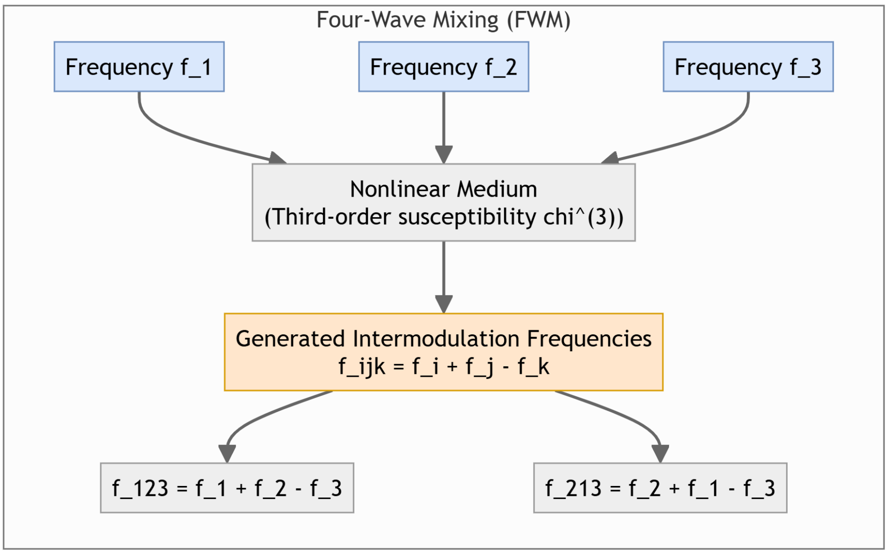

Figure 6.3: Four-Wave Mixing Sidebands

- **Example (Degenerate Case):** If f_1 and f_2 are two WDM channels, they generate sidebands at:

$$f_{FWM} = 2f_1 - f_2 \quad \text{and} \quad f_{FWM} = 2f_2 - f_1$$

- **Impact:** The generated FWM sidebands often fall directly on adjacent WDM channel frequencies, causing severe coherent crosstalk.
- **Mitigation:**
- **Non-Zero Dispersion-Shifted Fiber (NZ-DSF):** Introducing a small amount of local dispersion ($D \neq 0$ at the operating wavelength) causes the waves to travel at different velocities, breaking the phase-matching condition and suppressing FWM.
- **Unequal Channel Spacing:** Arranging WDM channels such that FWM products fall in the gaps between channels rather than directly on them.

6.4 4. Group Velocity Dispersion (GVD)

Group Velocity Dispersion (GVD) is the phenomenon where different frequency components of an optical pulse travel at different speeds, causing temporal pulse broadening.

6.4.1 4.1 Dispersion Parameter (D)

[5M][] Q6.2 (GVD Concept & Dispersion Parameter)

- **GVD Parameter (β_2):** Defined as the second derivative of the propagation constant β with respect to angular frequency ω:

$$\beta_2 = \frac{d^2\beta}{d\omega^2}$$

- **Dispersion Parameter (D):** Expresses dispersion in terms of wavelength (λ), with units of ps/(nm $\cdot$ km):

$$D = -\frac{2\pi c}{\lambda^2}\beta_2$$

6.4.2 4.2 Dispersion Regimes

Depending on the sign of β_2 and D, fibers operate in two distinct regimes:

1. **Normal Dispersion Regime ($\beta_2 > 0, D < 0$):**

- Occurs at wavelengths shorter than the zero-dispersion wavelength (e.g., $\lambda < 1310$ nm in standard silica fiber).
- Higher-frequency (blue-shifted) components travel **slower** than lower-frequency (red-shifted) components.

1. **Anomalous Dispersion Regime ($\beta_2 < 0, D > 0$):**

- Occurs at wavelengths longer than the zero-dispersion wavelength (e.g., $\lambda > 1310$ nm in standard silica fiber, including the 1550 nm window).
- Higher-frequency (blue-shifted) components travel **faster** than lower-frequency (red-shifted) components.

6.5 5. Soliton-based Communication

An optical soliton is a localized electromagnetic wave that propagates over long distances through a nonlinear dispersive medium while preserving its shape and velocity.

6.5.1 5.1 The Nonlinear Schrödinger Equation (NLSE)

[10M][] Q6.4 (a) (Soliton Concept & Balancing Mechanism)

The propagation of a pulse envelope $A(z,T)$ in a dispersive, nonlinear fiber is governed by the **Nonlinear Schrödinger Equation (NLSE)**:

$$j\frac{\partial A}{\partial z} - \frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2} + \gamma|A|^2 A = 0$$

Where:*

- $\frac{\partial^2 A}{\partial T^2}$ term models Group Velocity Dispersion (GVD).
- $|A|^2 A$ term models Self-Phase Modulation (SPM).

6.5.2 5.2 The Balancing Mechanism

A fundamental soliton ($N = 1$) exists only in the **anomalous dispersion regime** ($\beta_2 < 0$ or $D > 0$).

- **GVD Chirp:** In the anomalous regime, dispersion induces a frequency chirp where the leading edge of the pulse shifts to the blue side (travels faster) and the trailing edge shifts to the red side (travels slower).
- **SPM Chirp:** Self-Phase Modulation does the exact opposite: it induces a frequency chirp where the leading edge shifts to the red side and the trailing edge shifts to the blue side.
- **Cancellation:** By launching a pulse with a specific peak power and hyperbolic-secant shape, the blue-shifting GVD chirp and red-shifting SPM chirp cancel each other out. The pulse propagates indefinitely without temporal or spectral distortion.

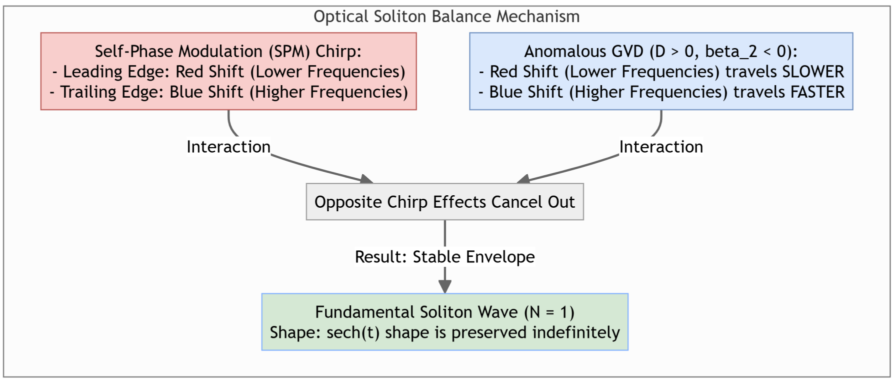

Figure 6.4: Soliton Propagation Balance

- **Pulse Shape of Fundamental Soliton:**

$$A(z,T) = \sqrt{P_0}\,\text{sech}\left(\frac{T}{T_0}\right) e^{jz/2L_D}$$

Where:*

- $P_0 = \frac{|\beta_2|}{\gamma T_0^2}$ is the peak power required.
- $L_D = \frac{T_0^2}{|\beta_2|}$ is the dispersion length.

6.5.3 5.3 System Advantages & Disadvantages

[10M][] Q6.4 (b) (Advantages & Disadvantages of Soliton Links)

Advantages

- **Zero Pulse Broadening:** Solitons maintain their shape over thousands of kilometers, eliminating intersymbol interference (ISI).
- **Ultra-High Bit Rates:** Soliton communication enables ultra-fast data rates (e.g., $>$ 40 Gbps per channel) over transoceanic distances.
- **Simple Dispersion Management:** Reduces the need for complex, lumped dispersion-compensating modules (DCM).

Disadvantages

- **Gordon-Haus Jitter:** Interaction between solitons and amplified spontaneous emission (ASE) noise from optical amplifiers causes random fluctuations in the soliton's carrier frequency, leading to arrival time jitter at the receiver.
- **Soliton Interactions:** Two closely spaced solitons can attract or repel each other depending on their relative phase, requiring larger channel guard bands and limiting the maximum channel capacity.
- **Soliton Self-Frequency Shift:** Intramodal Raman scattering transfers energy from higher-frequency to lower-frequency parts of the soliton spectrum, causing a continuous down-shift in carrier frequency.
- **High Transmitter Cost:** Requires highly stable, short-pulse source lasers (e.g., mode-locked lasers) and high-speed modulation electronics.

Chapter 7

Quick Revision: PE-EC801B - Fiber Optic Communication

This high-density revision sheet compiles core definitions, mathematical formulas, essential diagrams to practice drawing, frequently asked exam questions, and fast-recall points across all 6 units of the syllabus.

7.1 Important Definitions

7.1.1 Unit I: Introduction

- **Snell's Law:** A law stating that the ratio of the sines of the angles of incidence and refraction is equal to the ratio of velocities in the two media, or equivalently, the product of the refractive index and the sine of the angle is constant: $n_1 \sin\theta_1 = n_2 \sin\theta_2$.
- **Critical Angle (θ_c):** The minimum angle of incidence at the core-cladding boundary above which total internal reflection (TIR) occurs, defined as $\theta_c = \sin^{-1}(n_2/n_1)$.
- **Acceptance Angle (θ_a):** The maximum angle at which a ray of light entering the fiber core from air will be guided by total internal reflection along the fiber.
- **Numerical Aperture (NA):** A dimensionless parameter measuring the light-gathering ability of the fiber, defined as the sine of the acceptance angle: $\mathrm{NA} = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$.
- **Meridional Rays:** Rays of light that pass through the longitudinal axis of the fiber core, remaining within a single plane.
- **Skew Rays:** Rays that propagate through the fiber core without crossing the fiber's central longitudinal axis, following a helical path.

7.1.2 Unit II: Optical Fibers

- **Normalized Frequency (V-number):** A dimensionless parameter that determines the number of guided modes a fiber can support: $V = \frac{2\pi a}{\lambda}\mathrm{NA}$.
- **Cutoff Wavelength (λ_c):** The minimum wavelength at which a single-mode fiber supports only the fundamental mode (LP_{01}), defined under the condition $V = 2.405$.

- **Mode-Field Diameter (MFD):** A measure of the radial distribution of optical power in the fundamental mode of a single-mode fiber, representing the width where the intensity drops to $1/e^2$ of its peak.
- **Step-Index Fiber:** A fiber with a constant refractive index in the core and a sudden decrease (step) at the core-cladding boundary.
- **Graded-Index Fiber:** A fiber with a core refractive index that decreases continuously and parabolically from the center axis out to the cladding boundary.

7.1.3 Unit III: Signal Degradation

- **Attenuation:** The reduction in optical power of a signal as it propagates through a fiber, measured in dB/km.
- **Dispersion:** The temporal broadening of an optical pulse as it travels along a fiber, limiting the transmission bandwidth.
- **Intermodal Dispersion:** Pulse broadening in multimode fibers caused by different modes traveling at different group velocities, arriving at different times.
- **Intramodal (Chromatic) Dispersion:** Pulse broadening within a single mode caused by different wavelength components traveling at different speeds, consisting of material and waveguide dispersion.
- **Optical Time Domain Reflectometer (OTDR):** An instrument that injects high-power optical pulses into a fiber and measures the backscattered and reflected light (Rayleigh scattering and Fresnel reflection) to locate faults and measure attenuation.
- **GRIN-rod Lens:** A compact graded-index glass cylinder where the refractive index decreases parabolically from the central axis, used for collimating or focusing light.
- **Expanded-Beam Connector:** A fiber connector that uses lenses to expand and collimate the exiting beam across an air gap, reducing lateral alignment sensitivity.

7.1.4 Unit IV: Optical Sources & Detectors

- **Population Inversion:** A state in a laser medium where more electrons reside in a higher, excited energy state than in a lower, ground state, enabling net optical amplification.
- **Threshold Gain (g_{th}):** The minimum gain required in a laser cavity to balance the internal and mirror reflection losses, allowing laser oscillation to begin.
- **Responsivity (R):** The ratio of generated photocurrent to the incident optical power in a photodetector, measured in A/W.
- **Quantum Efficiency (η):** The ratio of the number of electron-hole pairs collected at the terminals of a photodetector to the number of incident photons.
- **Avalanche Photodiode (APD):** A highly sensitive photodetector that uses impact ionization under high reverse bias to provide internal current gain through avalanche multiplication.

- **Isotype & Anisotype Heterojunctions:** Isotype is a junction between two different materials of the same conductivity type (e.g. n-N). Anisotype is between different conductivity types (e.g. p-N).
- **Radiative Minority Carrier Lifetime (τ_r):** The average time an injected minority carrier survives in a band before undergoing radiative recombination.

7.1.5 Unit V: Optical Switches & Amplifiers

- **Erbium-Doped Fiber Amplifier (EDFA):** A lumped optical amplifier that uses Erbium ions embedded in a silica core, pumped at 980 nm or 1480 nm, to amplify signals directly in the 1550 nm window.
- **Raman Amplifier:** An amplifier that exploits Stimulated Raman Scattering (SRS) in the transmission fiber to transfer energy from high-power pump lasers to signal wavelengths distributed along the link.
- **Coupling Length (L_c):** The physical length of a directional coupler required for 100% of the optical power to transfer from the launch waveguide to the parallel waveguide.
- **Wavelength Division Multiplexing (WDM):** The technique of combining multiple optical carrier signals of different wavelengths onto a single optical fiber to increase data capacity.

7.1.6 Unit VI: Nonlinear Effects

- **Kerr Effect:** The phenomenon where the refractive index of a medium changes in proportion to the local optical intensity of the propagating light: $n(I) = n_0 + n_2 I$.
- **Self-Phase Modulation (SPM):** A nonlinear phase shift induced on an optical pulse by its own intensity envelope, leading to spectral broadening.
- **Four-Wave Mixing (FWM):** A parametric nonlinear process where three waves of different frequencies beat together to generate new intermodulation sidebands.
- **Optical Soliton:** A stable optical pulse that propagates over long distances without changing its shape, achieved by balancing anomalous Group Velocity Dispersion (GVD) and Self-Phase Modulation (SPM).

7.2 Key Formulas

7.2.1 1. Numerical Aperture (NA) & Acceptance Angle

$$\mathrm{NA} = \sin\theta_a = \sqrt{n_1^2 - n_2^2} \approx n_1\sqrt{2\Delta}$$

$$\Delta = \frac{n_1 - n_2}{n_1}$$

- Where:
- θ_a = Acceptance angle in air

- n_1 = Core refractive index
- n_2 = Cladding refractive index
- Δ = Relative refractive index difference

Numerical Example:* Given $n_1 = 1.48$ and $n_2 = 1.46$:

$$\Delta = \frac{1.48 - 1.46}{1.48} \approx 0.0135$$

$$\text{NA} = \sqrt{1.48^2 - 1.46^2} = \sqrt{2.1904 - 2.1316} \approx \mathbf{0.242}$$

$$\theta_a = \sin^{-1}(0.242) \approx \mathbf{14.0}^\circ$$

7.2.2 2. Normalized Frequency (V-number) & Mode Count (N)

$$V = \frac{2\pi a}{\lambda}\text{NA}$$

$$N_{\text{step}} \approx \frac{V^2}{2} \quad \text{and} \quad N_{\text{graded}} \approx \frac{V^2}{4}$$

- Where:
- a = Core radius
- λ = Operating wavelength
- $N_{\text{step}}, N_{\text{graded}}$ = Total number of guided modes

Numerical Example:* A step-index fiber with core diameter 80 μm ($a = 40$ μm), NA $= 0.2$, operating at $\lambda = 0.85$ μm:

$$V = \frac{2\pi \times 40}{0.85} \times 0.2 \approx \mathbf{59.14}$$

$$N_{\text{step}} \approx \frac{59.14^2}{2} \approx \mathbf{1749} \text{ modes}$$

7.2.3 3. Cutoff Wavelength (λ_c) for Single-Mode Fiber

$$\lambda_c = \frac{2\pi a \text{NA}}{V_c} = \frac{2\pi a \text{NA}}{2.405}$$

- Where:
- $V_c = 2.405$ is the single-mode cutoff limit.

Numerical Example:* For core radius $a = 4.5$ μm and NA $= 0.12$:

$$\lambda_c = \frac{2\pi \times 4.5 \times 0.12}{2.405} \approx \mathbf{1.409}\ \mu\text{m} = \mathbf{1409} \text{ nm}$$

7.2.4 4. Attenuation Coefficient (α)

$$\alpha_{\text{dB/km}} = \frac{10}{L}\log_{10}\left(\frac{P_{\text{in}}}{P_{\text{out}}}\right)$$

- Where:
- L = Fiber link length in km
- P_{in} = Optical power launched at input
- P_{out} = Optical power measured at output

7.2.5 5. Multimode Intermodal Delay ($\Delta\tau$) & RMS Broadening (σ_{inter})

$$\Delta\tau = \frac{Ln_1\Delta}{c} \quad \text{and} \quad \sigma_{\text{inter}} \approx \frac{Ln_1\Delta}{2\sqrt{3}c}$$

- Where:
- c = Speed of light in vacuum ($\approx 3\times10^5$ km/s)
- $\Delta\tau$ = Total delay difference between fastest and slowest modes.

7.2.6 6. Photodetector Responsivity (R) & Quantum Efficiency (η)

$$R = \frac{\eta q}{h\nu} = \frac{\eta q\lambda}{hc} \approx \frac{\eta\lambda}{1.24}$$

- Where:
- q = Electron charge (1.6×10^{-19} C)
- h = Planck's constant (6.626×10^{-34} J $\cdot$ s)
- λ = Wavelength in μm

Numerical Example:* A photodiode operating at $\lambda = 1.3$ μm with quantum efficiency $\eta =$ 70% (0.7):

$$R \approx \frac{0.7\times1.3}{1.24} \approx \mathbf{0.734} \text{ A/W}$$

7.2.7 7. Laser Cavity Dynamics and Physics Equations

- **Facet Reflectivity (R):**

$$R = \left(\frac{n-1}{n+1}\right)^2$$

(where n is refractive index of active semiconductor).

- **Longitudinal Mode Count (m):**

$$m = \frac{2nL}{\lambda}$$

(where L is cavity length and λ is free-space wavelength).

- **Mode Frequency Separation (Δf):**

$$\Delta f = \frac{c}{2nL}$$

- **Mode Wavelength Separation ($\Delta\lambda$):**

$$\Delta\lambda = \frac{\lambda^2 \Delta f}{c} = \frac{\lambda^2}{2nL}$$

- **Radiative Minority Carrier Lifetime (τ_r):**

$$\tau_r = \frac{1}{B \cdot p_0}$$

(where B is recombination coefficient and p_0 is majority carrier hole concentration).

7.2.8 7. Rise-Time Budget for Digital Links

$$t_{\text{sys}} = \sqrt{t_{\text{tx}}^2 + t_{\text{disp}}^2 + t_{\text{rx}}^2} \leq 0.7T \quad \text{(for NRZ coding)}$$

- Where:
- t_{sys} = Total system rise-time.
- $t_{\text{tx}}, t_{\text{rx}}$ = Transmitter and receiver electrical rise-times.
- t_{disp} = Fiber dispersion rise-time (chromatic + modal).
- T = Bit period ($1/\text{BitRate}$).

7.2.9 8. Directional Coupler Power Equations

$$P_1(z) = P_0 \cos^2(\kappa z) \quad \text{and} \quad P_2(z) = P_0 \sin^2(\kappa z)$$

$$L_c = \frac{\pi}{2\kappa}$$

- Where:
- κ = Evanescent coupling coefficient
- L_c = Coupling length

7.2.10 9. Soliton Peak Power (P_0)

$$P_0 = \frac{|\beta_2|}{\gamma T_0^2}$$

- Where:
- β_2 = Group Velocity Dispersion (GVD) parameter (s^2/m)
- γ = Fiber nonlinearity coefficient ($\text{W}^{-1}\text{km}^{-1}$)
- T_0 = Half-width (pulse duration parameter) of the soliton

7.3 Diagrams to Practice Drawing

Prepare to sketch these block architectures, ray paths, and schematic layouts by hand for long-form answers:

7.3.1 1. General & Fiber-Optic Communication Systems (Unit I)

Detailed comparison of general electronic vs. photonic transmission pipelines.

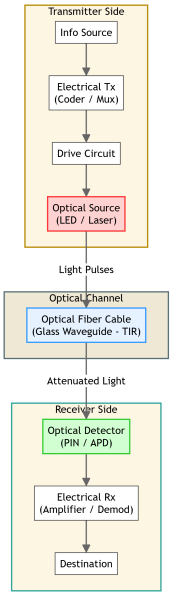

Figure 7.1: General & Fiber-Optic Communication System

7.3.2 2. Ray Propagation & Numerical Aperture (Unit I)

Ray paths at the core-cladding boundary showing refraction, critical angle, and acceptance cone.

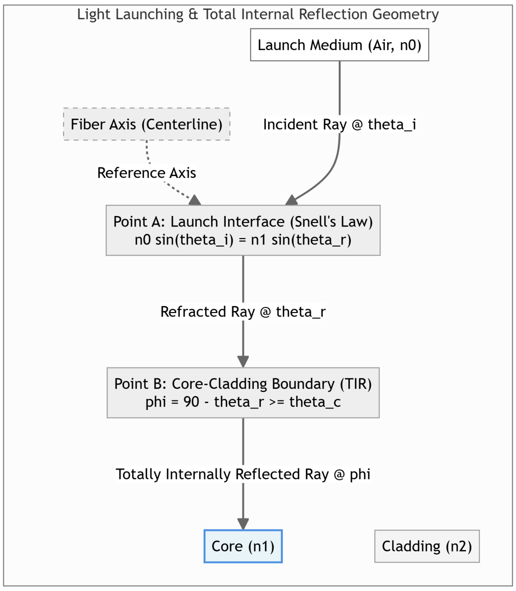

Figure 7.2: Ray Propagation & Angles

7.3.3 3. Attenuation Curve & Transmission Windows (Unit III)

Wavelength dependent attenuation showing Rayleigh scattering, infrared absorption, and the 1310/1550 nm windows.

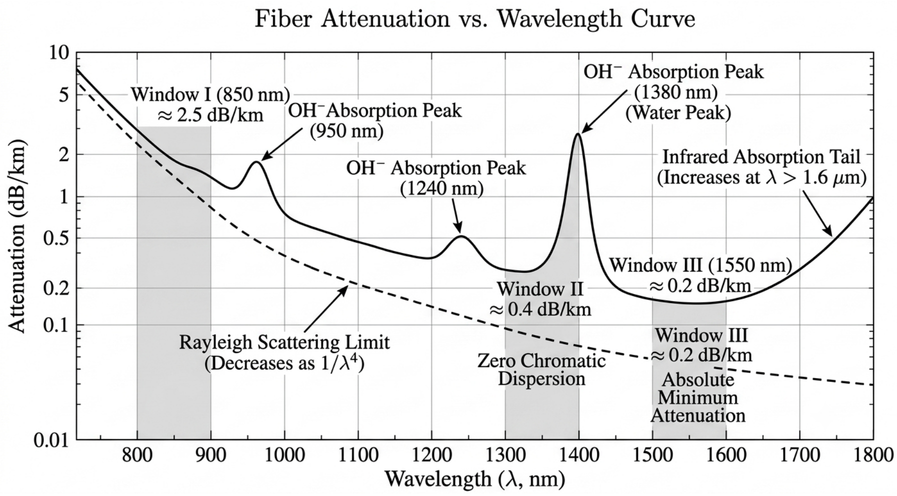

Figure 7.3: Attenuation Curve & Transmission Windows

7.3.4 4. Multimode vs. Single-Mode Pulse Broadening (Unit III)

Illustration of how multiple paths lead to dispersion and overlapping output pulses.

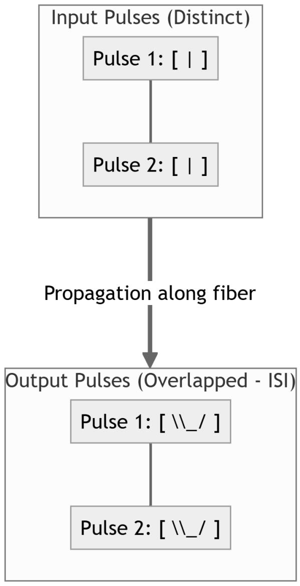

Figure 7.4: Pulse Broadening

7.3.5 5. OTDR Configuration & Trace (Unit III)

OTDR optical splitter block diagram and the corresponding backscatter log trace identifying splices, faults, and reflection peaks.

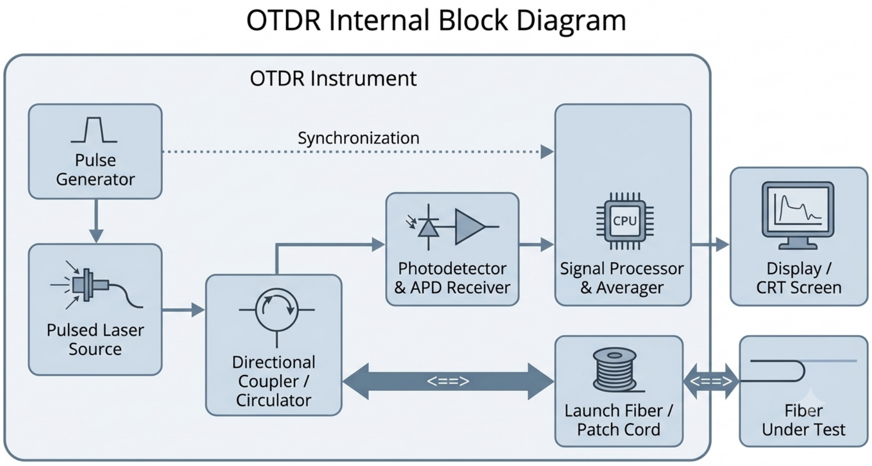

Figure 7.5: OTDR block diagram

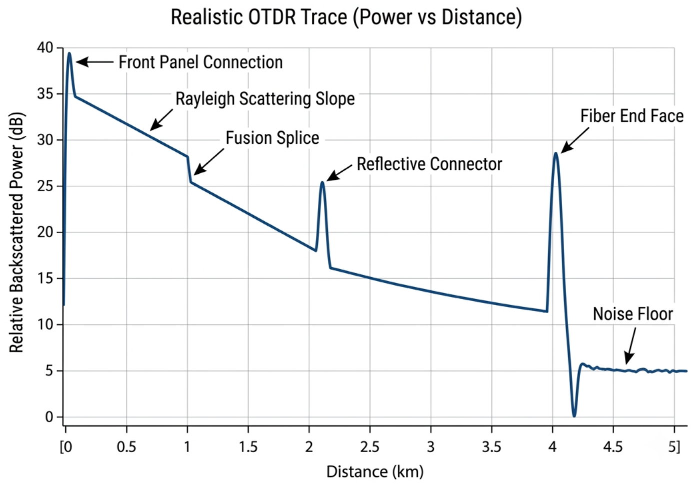

Figure 7.6: OTDR trace

7.3.6 6. Laser Resonator Cavity (Unit IV)

Double-heterostructure gain medium with cleaved mirror end-facets highlighting optical feedback.

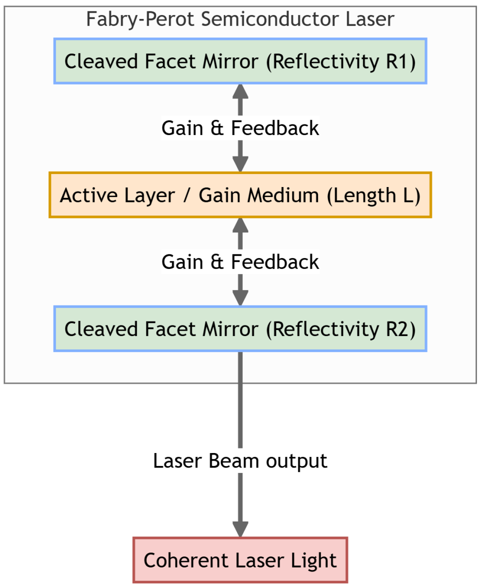

Figure 7.7: Laser Cavity structure

7.3.7 7. Digital Optical Receiver Pipeline (Unit IV)

Block diagram tracking signal from photodiode through pre-amplifier, filter, and decision circuit.

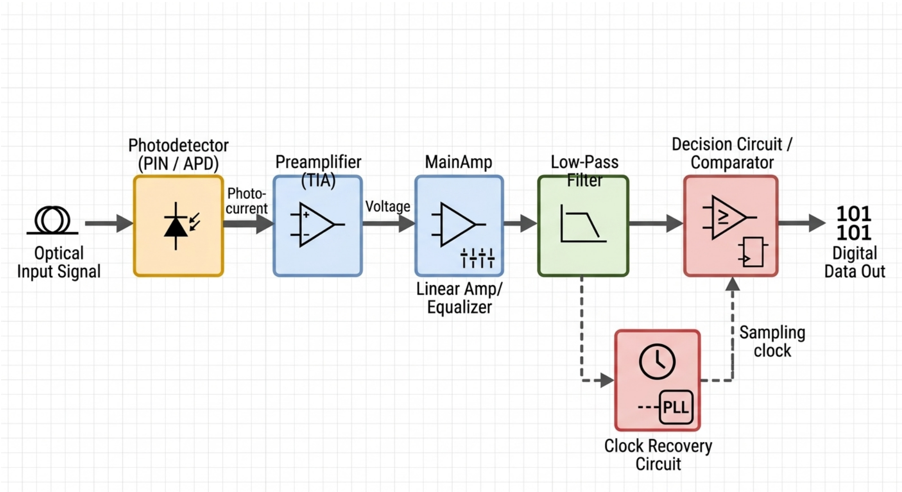

Figure 7.8: Optical Receiver structure

7.3.8 8. EDFA Physical Layout (Unit V)

Optical configuration detailing Erbium spool, pump laser, WDM coupler, and input/output isolators.

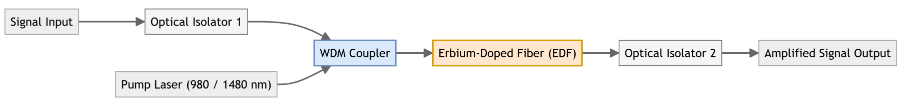

Figure 7.9: EDFA Block diagram

7.3.9 9. Directional Coupler States (Unit V)

Layout showing coupling paths for Cross State (zero volts) and Bar State (voltage applied).

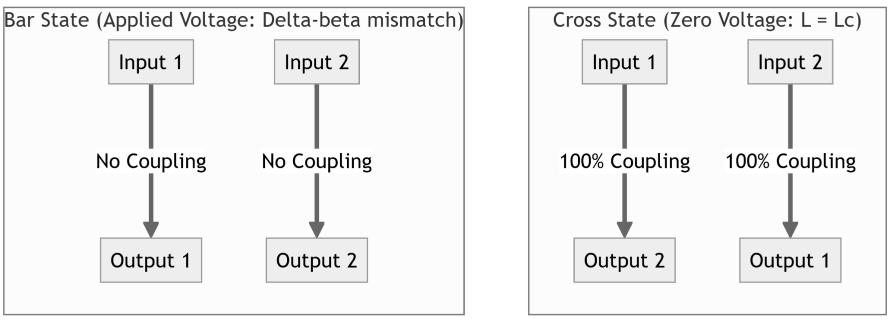

Figure 7.10: Directional Coupler Switch States

7.3.10 10. MZI Optical Switch Layout (Unit V)

Y-splitter arms with deposited metal electrodes creating phase difference via electro-optic index shift.

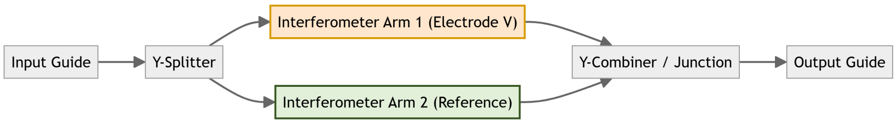

Figure 7.11: MZI Switch Layout

7.3.11 11. OADM and OXC Routing Architectures (Unit V)

Wavelength drop/add filters and cross-connect fiber routing tables.

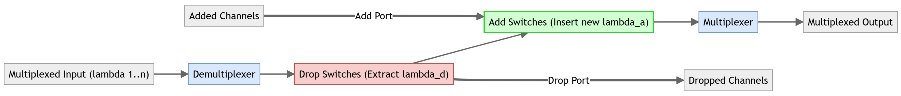

Figure 7.12: OADM Architecture

Figure 7.13: OXC Wavelength Routing

7.3.12 12. SRS vs. SBS Scattering Processes (Unit VI)

Energy transition flowcharts comparing optical molecular phonons (SRS) vs. acoustic density wave phonons (SBS).

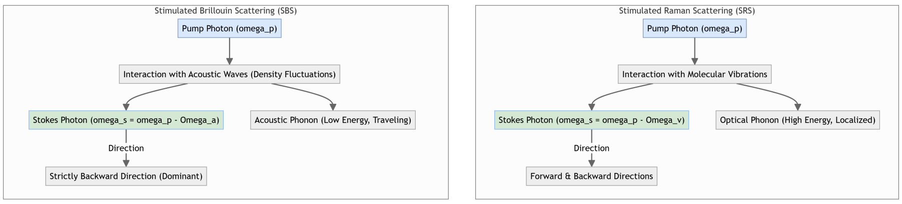

Figure 7.14: SRS vs. SBS scattering

7.3.13 13. SPM Pulse Chirping (Unit VI)

Envelope index changes mapping leading edge red shifts and trailing edge blue shifts.

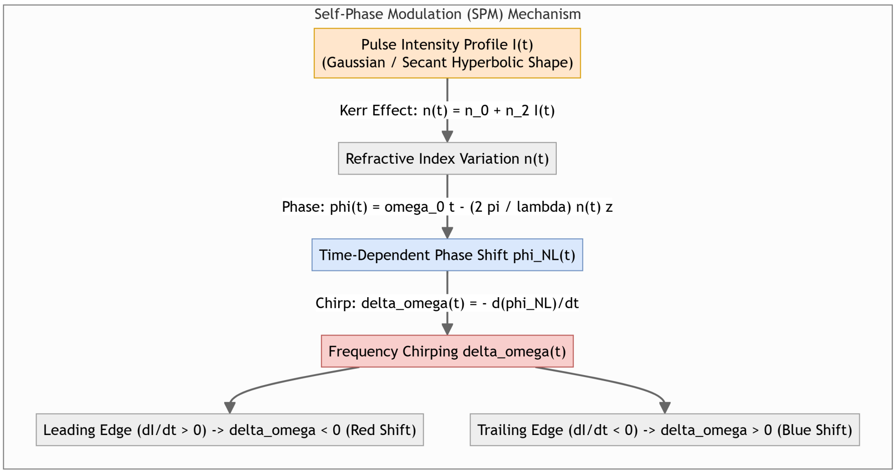

Figure 7.15: SPM frequency chirping

7.3.14 14. Soliton Envelope Balance (Unit VI)

Interaction curve showing anomalous GVD compression exactly counteracting SPM broadening.

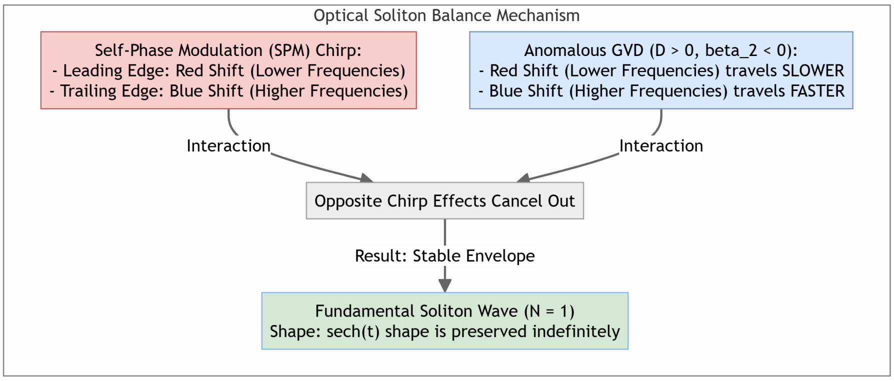

Figure 7.16: Soliton propagation balance

7.4 Frequently Asked Questions

7.4.1 Q1. Derive the expression for Acceptance Angle and Numerical Aperture (NA) of a multimode step-index fiber. [10M]

- **Step 1:** Consider a ray of light entering the fiber core from a medium of refractive index n_0 (air, $n_0 = 1$) at an angle θ_0 to the fiber axis.
- **Step 2:** Apply Snell's Law at the air-core interface:

$$n_0 \sin\theta_0 = n_1 \sin\theta_1$$

- **Step 3:** The refracted ray strikes the core-cladding boundary at an angle of incidence ϕ. For total internal reflection to occur, $\phi \geq \theta_c$. From geometry:

$$\theta_1 = 90^\circ - \phi \implies \sin\theta_1 = \cos\phi$$

- **Step 4:** Substitute this into Snell's Law:

$$n_0 \sin\theta_0 = n_1 \cos\phi = n_1\sqrt{1-\sin^2\phi}$$

- **Step 5:** Under critical conditions, $\sin\phi = \sin\theta_c = n_2/n_1$. Substituting this yields:

$$n_0 \sin\theta_a = n_1\sqrt{1-\left(\frac{n_2}{n_1}\right)^2} = \sqrt{n_1^2 - n_2^2}$$

- **Step 6:** Setting $n_0 = 1$ for air, we get the final expression for NA:

$$\mathrm{NA} = \sin\theta_a = \sqrt{n_1^2 - n_2^2}$$

7.4.2 Q2. Compare step-index and graded-index fibers in terms of profiles, ray paths, and intermodal dispersion. [5M]

- **Refractive Index Profile:** Step-index has a constant index n_1 in the core with a step decrease at the cladding. Graded-index has a parabolic index profile decreasing from the core axis outward.
- **Ray Propagation Path:** Light travels in zig-zag straight lines in step-index cores. In graded-index cores, rays bend continuously in sinusoidal, helical paths.
- **Intermodal Dispersion:** High in step-index fibers because rays traveling at larger angles take longer paths. Low in graded-index fibers because rays traveling on longer outer paths travel faster (where refractive index is lower), keeping arrival times synchronized.

7.4.3 Q3. Derive the intermodal delay difference ($\Delta\tau$) in a step-index fiber. [10M]

- **Fastest Ray:** Propagates along the central longitudinal axis of length L. Travel time is:

$$t_{\min} = \frac{L}{v} = \frac{Ln_1}{c}$$

- **Slowest Ray:** Travels at the critical angle θ_c, reflecting repeatedly. Path length is $L/\sin\theta_c = L(n_1/n_2)$. Travel time is:

$$t_{\max} = \frac{L(n_1/n_2)}{v} = \frac{Ln_1^2}{cn_2}$$

- **Delay Difference ($\Delta\tau$):**

$$\Delta\tau = t_{\max} - t_{\min} = \frac{Ln_1}{c}\left(\frac{n_1}{n_2} - 1\right) = \frac{Ln_1}{c}\left(\frac{n_1 - n_2}{n_2}\right)$$

- Since $n_2 \approx n_1$ for weakly guiding fibers:

$$\Delta\tau \approx \frac{Ln_1\Delta}{c}$$

7.4.4 Q4. Detail the differences between PIN photodiodes and Avalanche Photodiodes (APDs). [5M]

- **Structure:** PIN photodiode has a standard Intrinsic semiconductor layer between P and N regions. APD has an additional high-field multiplication region.
- **Gain Mechanism:** PIN photodiodes have no internal gain (one electron per photon, $M = 1$). APDs provide internal gain ($M = 10$ to 100) through avalanche multiplication.
- **Responsivity:** PIN responsibility is lower (≈ 0.6 to 0.8 A/W). APD responsivity is much higher due to multiplication factor M.
- **Noise & Bias:** PIN photodiodes require low reverse bias voltage (5 to 15 V) and generate low noise. APDs require very high reverse bias voltage (100 to 300 V) and introduce excess avalanche noise.

7.4.5 Q5. Differentiate between Erbium-Doped Fiber Amplifiers (EDFA) and Raman Amplifiers. [5M]

- **Gain Medium:** EDFA uses a short, lumped section of silica fiber heavily doped with Erbium ions. Raman amplification uses the standard long transmission fiber itself (distributed gain).
- **Amplification Principle:** EDFA uses stimulated emission from excited Erbium ions. Raman uses Stimulated Raman Scattering (SRS) where high-power pump photons transfer energy to signal photons.
- **Operating Band:** EDFA is restricted to the fixed C-band (1530 to 1565 nm) and L-band (1565 to 1625 nm). Raman amplifiers can operate at any wavelength by tuning the pump laser frequency.
- **Pump Power:** EDFAs require moderate pump powers (10 to 100 mW). Raman amplifiers require very high pump powers (> 500 mW).

7.4.6 Q6. Explain how optical solitons propagate without broadening in a fiber. [10M]

- **The Problem:** Normal pulse propagation suffers from Group Velocity Dispersion (GVD), which spreads different frequency components temporally, causing pulse broadening.
- **Anomalous GVD:** At wavelengths > 1310 nm in standard fiber, GVD is anomalous ($D > 0$), meaning higher-frequency (blue) components travel faster than lower-frequency (red) components, creating a GVD chirp (blue leading, red trailing).
- **SPM Effect:** Due to the intensity-dependent Kerr effect, Self-Phase Modulation (SPM) creates a nonlinear phase shift that shifts the leading edge to lower frequencies (red shift) and the trailing edge to higher frequencies (blue shift).
- **The Balance:** Because the chirps introduced by anomalous GVD and SPM are equal and opposite, they cancel each other out. The optical pulse propagates as a fundamental soliton ($N = 1$), preserving its shape indefinitely.

7.5 One-Line Revision Bullets

- **Unit I:** Total internal reflection requires light to travel from a denser core (n_1) to a rarer cladding (n_2) at an angle greater than the critical angle θ_c.
- **Unit II:** Single-mode propagation in a step-index fiber is achieved only when the normalized frequency $V < 2.405$.
- **Unit III:** Intramodal dispersion consists of material dispersion (wavelength-dependent index) and waveguide dispersion (wavelength-dependent boundary guidance), and it dominates in single-mode links.
- **Unit IV:** Laser diodes rely on optical feedback in a resonator cavity to achieve stimulated emission, whereas LEDs rely on spontaneous emission.
- **Unit V:** Mach-Zehnder Interferometers act as high-speed optical switches by applying voltage across electrodes to induce a π phase difference between interferometer arms.
- **Unit VI:** Four-Wave Mixing generates severe crosstalk in WDM systems, which is mitigated by introducing local chromatic dispersion (NZ-DSF) to break phase-matching.